

A Systematic Review of Metal Halide Perovskite Crystallization and Film Formation Mechanism Unveiled by In Situ GIWAXS

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Metal halide perovskites are of fundamental interest in the research of modern thin-film optoelectronic devices, owing to their widely tunable optoelectronic properties and solution processability. To obtain high-quality perovskite films and ultimately high-performance perovskite devices, it is crucial to understand the film formation mechanisms, which, however, remains a great challenge, due to the complexity of perovskite composition, dimensionality, and processing conditions. Nevertheless, the state-of-the-art in situ grazing-incidence wide-angle X-ray scattering (GIWAXS) technique enables one to bridge the complex information with device performance by revealing the crystallization pathways during the perovskite film formation process. In this review, the authors illustrate how to obtain and understand in situ GIWAXS data, summarize and assess recent results of in situ GIWAXS studies on versatile perovskite photovoltaic systems, aiming at elucidating the distinct features and common ground of film formation mechanisms, and shedding light on future opportunities of employing in situ GIWAXS to study the fundamental working mechanisms of highly efficient and stable perovskite solar cells toward mass production.

1. Introduction

Metal halide perovskites, named after their crystal structure with a general formula of ABX_3 (A = methylammonium (MA^+), formamidinium (FA^+), or Cs^+ ; B = Pb^{2+} , Sn^{2+} , or Ge^{2+} ; X = I^- , Br^- , or Cl^-), have attracted extensive attention as one of the most promising materials for applications in different optoelectronic devices,^[1] owing to their widely tunable optoelectronic properties and solution processability. The diverse optoelectronic properties, such as bandgap,^[2] absorption,^[3] electroluminescence,^[4] exciton diffusion,^[5] and charge transport properties,^[6] are closely related to the perovskite composition, crystal structure, and film quality. The latter two aspects are highly sensitive to the film fabrication process. Therefore, a thorough understanding of the perovskite crystallization

kinetics and film formation mechanism is greatly desirable, yet, very challenging, due to the complexity of perovskite composition, dimensionality, and processing conditions.

In terms of composition and dimensionality, perovskite systems can be divided into many types: three-dimensional (3D) organic–inorganic perovskites,^[7] two-dimensional (2D) perovskites,^[8] all-inorganic perovskites,^[9] and lead-free perovskites,^[10] which in turn diversify the crystallization pathways. The formation of 3D organic–inorganic perovskites usually evolves from intermediate phases to the cubic or tetragonal phase,^[11] whereas the crystallization of 2D perovskites often exhibits a direct phase transition from the amorphous precursor to the orthorhombic phase.^[12] All-inorganic perovskites, in contrast, present temperature-dependent phase transitions between cubic α -phase, tetragonal β -phase, orthorhombic γ -phase, and orthorhombic δ -phase.^[13] Furthermore, processing conditions can also substantially affect or alter the perovskite phase transition and crystallization processes. For instance, the one-step spin-coating method can lead to a complex phase competition between the intermediate phase and perovskite phase,^[14] while the two-step method will result in a different phase transition pathway from the hexagonal PbI_2 phase to the perovskite phase.^[15] In addition, additives are reported to either accelerate or retard the crystallization speed.^[16] **Figure 1** summarizes the commonly used organic cations for 3D and 2D perovskites and versatile perovskite, secondary and intermediate crystal structures that could involve in perovskite formation processes.

To monitor perovskite crystallization and film formation process and to establish the correlation between perovskite system, processing condition, material optoelectronic property, and device performance, the state-of-the-art synchrotron-based in situ grazing-incidence wide-angle X-ray scattering (GIWAXS) technique is usually employed.^[17] GIWAXS is a conventional scattering technique to probe the crystal structure of thin films. Equipped with high-brilliance synchrotron radiation and suitable sample environments, one can conduct in situ GIWAXS measurements with a very short exposure time to monitor the real-time structural evolution and crystallization pathways even for organic materials with low electron density.^[18] It is noted

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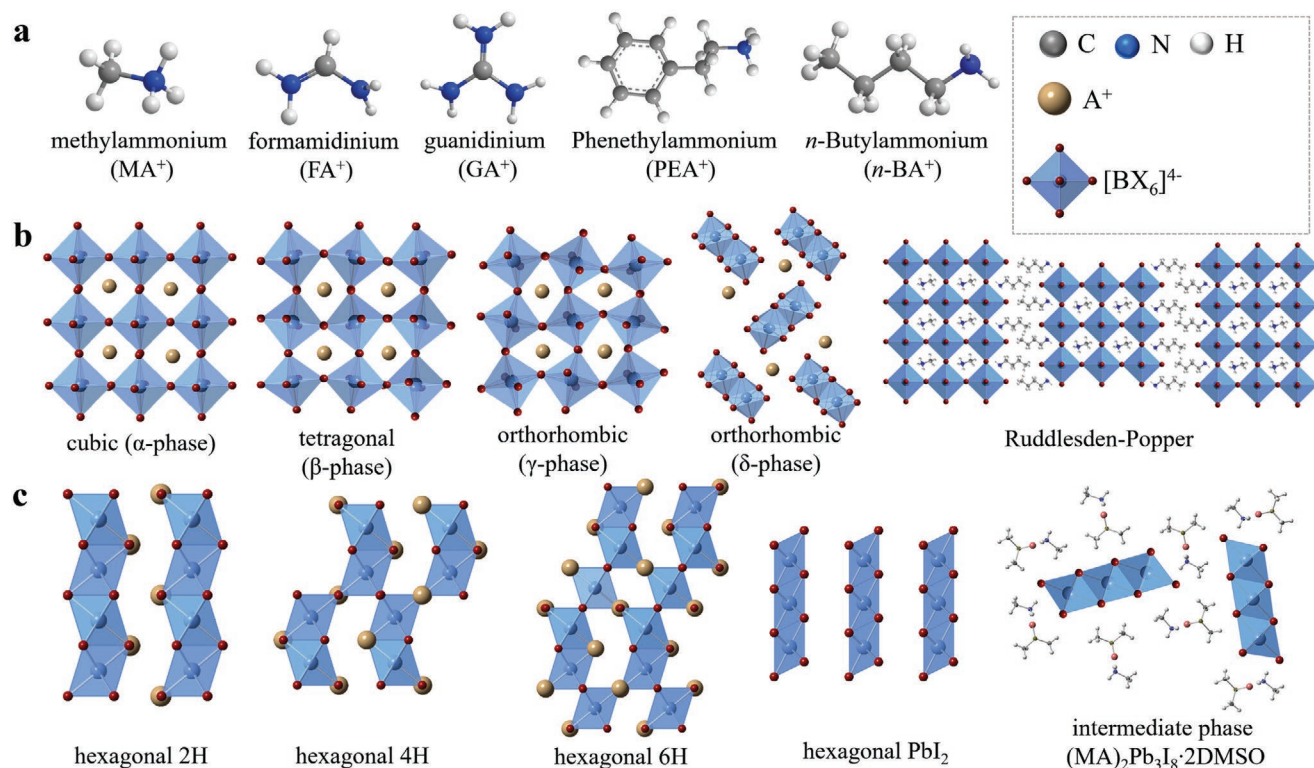


Figure 1. a) Commonly used organic cations, b) versatile perovskite phases, and c) secondary/intermediate phases involved in perovskite formation processes.

that the incident X-ray beam should be carefully attenuated in order to avoid the beam damage on materials. And it is feasible to conduct in situ GIWAXS on perovskite materials using an in-house X-ray source due to the high scattering cross-section of perovskites.^[19] Consequently, this technique is gradually becoming an indispensable tool to investigate the film formation mechanism of perovskites.

In this review, we first give a brief introduction of the in situ GIWAXS characterization and its application to photovoltaic materials. Then the perovskite structural evolution and crystallization processes studied by in situ GIWAXS are systematically reviewed based on different perovskite compositions in the sequence of lead halide perovskites and lead-free perovskites. Lead halide perovskites are further divided into MA-based perovskites, FA-based perovskites, mixed perovskites, Cs-based all-inorganic perovskites, and long-chain ammonium incorporated 2D perovskites. The impacts of different deposition methods, as well as various film treatment procedures on the perovskite crystallization pathways, are also included. Moreover, the correlations between the film formation process, perovskite systems, film properties, and device performance are presented to comprehensively understand the perovskite film formation mechanisms. Finally, potential future research directions of in situ GIWAXS on perovskite materials are discussed. This review summarizes in-depth understandings of the perovskite film formation mechanisms and insightful guidelines to improve the fabrication of efficient PSCs through crystallization manipulations.

2. In Situ GIWAXS Characterization

GIWAXS is one of the most popular grazing-incidence X-ray scattering (GIXS) techniques to probe the microstructure of thin films. The measuring configuration of a GIXS experiment is illustrated in **Figure 2**: an X-ray beam with the initial wave vector $\vec{k}_i$ shines on the sample surface at a grazing angle of α_i , and the scattered wave $\vec{k}_f$ is collected by an area detector. The scattering wave vector $\vec{q} = \vec{k}_f - \vec{k}_i$, can be expressed in the following components:^[21]

$$q_x = \frac{2\pi}{\lambda} (\cos \alpha_f \cos \varphi - \cos \alpha_i) \quad (1)$$

$$q_y = \frac{2\pi}{\lambda} (\cos \alpha_f \sin \varphi) \quad (2)$$

$$q_r = \sqrt{q_x^2 + q_y^2} \quad (3)$$

$$q_z = \frac{2\pi}{\lambda} (\sin \alpha_i + \sin \alpha_f) \quad (4)$$

where $\vec{q}_z$ is along the out-of-plane (OOP) direction with respect to the sample surface, $\vec{q}_x$ and $\vec{q}_y$ are both in the in-plane (IP) direction, being parallel and perpendicular to the incident plane, respectively. It is worth mentioning that there exists an inaccessible region or so-called “missing wedge” in the symbolic GIWAXS pattern in Figure 1, owing to the curvature of the Ewald sphere.^[20] As a consequence, diffraction signals from

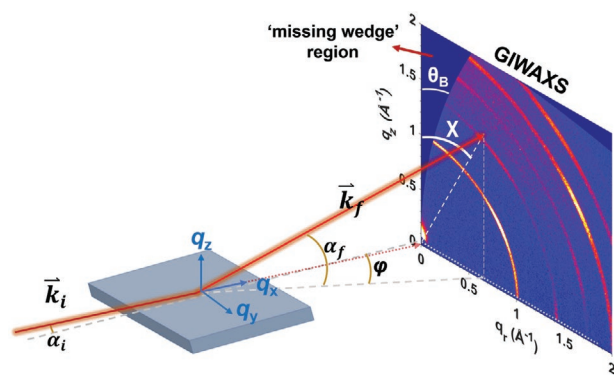


Figure 2. Scattering geometry of GIWAXS. The incident beam of the wave vector $\vec{k}_i$ hits the sample with a shallow incident angle of α_i , producing a scattered wave $\vec{k}_f$ with an outgoing angle α_f with respect to the sample surface. The outgoing plane has an azimuthal angle φ with respect to the incident beam plane. The direct beam points to the origin of the reciprocal space. The polar angle χ is the angle between the q_z axis and the projection of the scattering wave vector $\vec{q}$ on the q_r - q_z plane, while the polar angle θ_B corresponds to the minimum detectable χ .^[20]

highly ordered crystallites may be hidden in the inaccessible zone. Conventional co-planar X-ray diffraction (XRD) to probe the specular reflection along the q_z axis can be measured to solve this issue.^[20]

Another advantage of the GIWAXS characterization is the 2D area detector which can acquire a simultaneous signal collection in both the IP and the OOP directions. Hence, it provides additional orientational order of perovskite polycrystalline films. For spin-coated thin films which are supposed to

be isotropic in the IP direction, $\vec{q}_r = \vec{q}_x + \vec{q}_y$ is plotted versus $\vec{q}_z$ in 2D GIWAXS patterns. For a better illustration, a typical GIWAXS pattern of a highly oriented perovskite film is presented in **Figure 3**. The discrete spots at $q = 1.0 \text{ \AA}^{-1}$ along q_z and q_r axes can be ascribed to the lattice planes that are parallel and perpendicular to the surface, respectively, while both of them exhibit the same d -spacing value of $d = 2\pi/q = 6.3 \text{ \AA}$. It is noted that an additional peak at $q = 1.0 \text{ \AA}^{-1}$ with a χ value of 45° is detected, indicative of the existence of another grain orientation in the film,^[17] which further highlights the advantage of the GIWAXS characterization that all sets of grain orientation can be revealed in one GIWAXS pattern. As for the discrete diffraction spot located at $q = 1.4 \text{ \AA}^{-1}$ with a χ value of 45° , it corresponds to the lattice planes with a d -spacing value of $4.4 \text{ \AA}$ and an inclined angle of 45° with respect to the surface. In the same way, another observed peak at $q = 1.7 \text{ \AA}^{-1}$ with a χ value of 54.7° comes from the lattice planes with a d -spacing value of $3.6 \text{ \AA}$ and an inclined angle of 54.7° . Moreover, crystallite orientation in thin-film materials can be precisely quantified via GIWAXS using the mosaicity factor (MF)^[22] or the Hermans' orientation function (HF).^[23]

In addition, the depth profile of the crystal structures in perovskite films can be readily accessed through GIWAXS measurements by adjusting the incident angle of the incident beam to control the X-ray penetration depth, as shown in **Figure 4a**. When the incident angle α_i is smaller than the critical angle α_c , a total external reflection of the X-ray beam occurs, and thus only surface structural information of a few nanometers in thickness of the film can be probed. Raising the incident angle close to or slightly above the critical angle, one

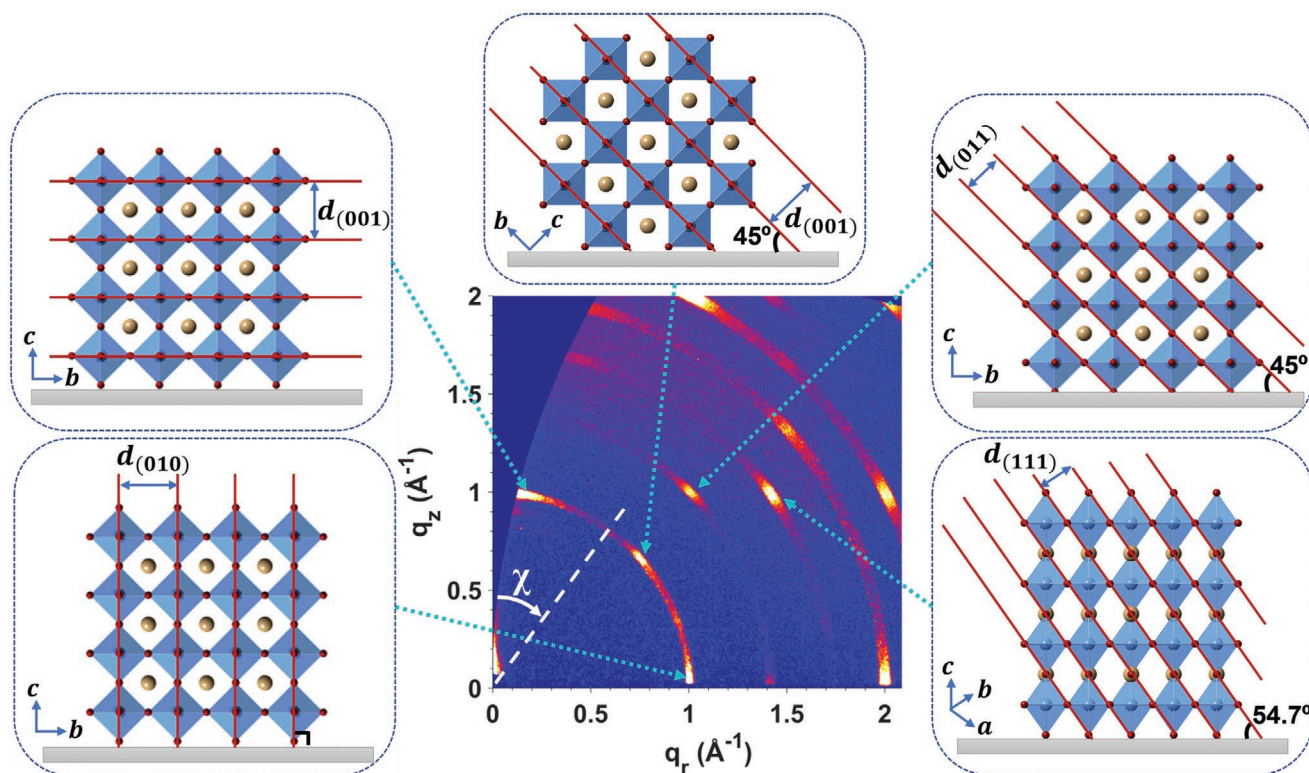


Figure 3. A schematic illustration of the GIWAXS pattern obtained from a highly oriented 3D perovskite film.

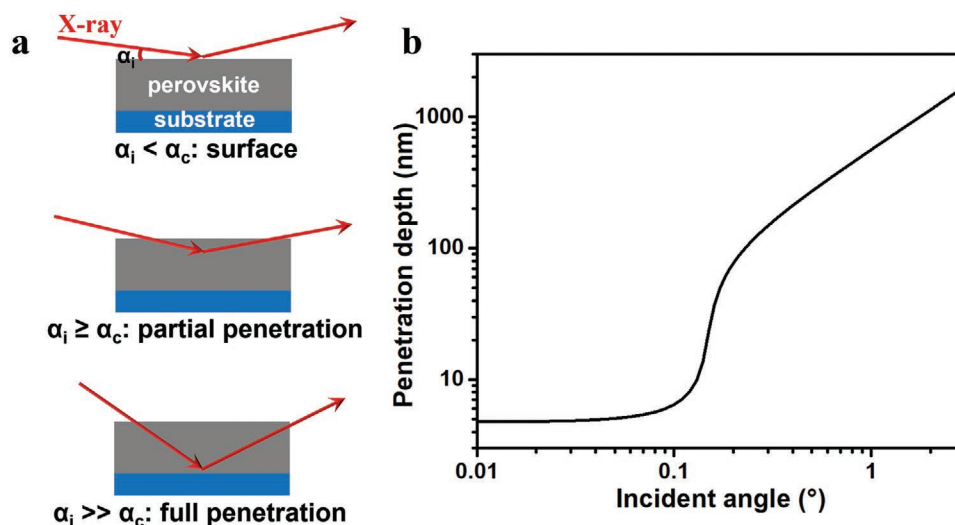


Figure 4. a) Schematics of the dependence of X-ray penetration depth on the incident angle α_i and the critical angle α_c . b) The estimated X-ray penetration depth in a typical 3D perovskite film; the X-ray wavelength is 1.54 Å (cf. Supporting Information of ref. [15]).

can measure the crystal structure of the film within a certain depth owing to the partial penetration of X-rays. A full penetration of X-rays into the bulk film can be achieved by progressively increasing the incident angle until the observation of substrate signals, which is in turn probing the average structural information throughout the whole film. Take the perovskite film as an example, the corresponding critical angle of the film probed by an X-ray of certain energy can be estimated based on the incident angle barely observing the substrate signal and the thickness of the film. Figure 4b plots the simulated penetration depth versus different incident angles in a typical polycrystalline 3D perovskite film with a critical angle of 0.15° at 10 keV.^[15] Below the critical angle, the X-rays only penetrate the surface of the film with several nanometers, whereas a much larger incident angle such as 1° enables the full penetration of the X-rays into the film. Therefore, one can perform GIWAXS measurements at different incident angles to investigate the uniformity of the bulk film along the vertical direction.^[24]

Remarkably, coupled with high flux and brilliance synchrotron radiation X-ray beam, one can measure bright GIWAXS patterns of perovskite films with millisecond exposure, thus enabling a real-time observation of the fast crystallization and phase transition processes of perovskites. The in situ GIWAXS characterization has been widely applied to monitor manifold perovskite deposition procedures such as spinning, thermal annealing, slot-die printing, and blade-coating processes (Figure 5a–e). In the meantime, an airtight chamber allows researchers to manipulate the perovskite fabrication environment (e.g., N₂, O₂, moisture, and light) and hence examine their influences on the film formation pathways. Furthermore, in operando IV tests combined with in situ GIWAXS measurements can directly correlate the perovskite structural evolutions with the device performance (Figure 5f), guiding the optimization of perovskite films and device fabrication. In short, in situ GIWAXS characterization is emerging as a powerful means to investigate the perovskite crystallization kinetics and film formation mechanisms under various processing conditions.

3. Crystallization Kinetics of Lead Halide Perovskites

In this section, we will systematically review the crystallization kinetics of lead halide perovskites based on various A-site cations including MA-based perovskites, FA-based perovskites, mixed perovskites, Cs-based all-inorganic perovskites, and long-chain ammonium incorporated 2D perovskites. Apart from the various A-site cations, different deposition procedures can also lead to diverse perovskite crystallization pathways. A perovskite film is generally deposited through two major steps: forming an as-cast film by spin coating and thermal annealing, which results in successive crystallization processes. In addition, different deposition procedures such as one-step spin coating, two-step spin coating, and blade coating also go through diverse crystallization pathways, resulting in distinct film properties. Accordingly, previous in situ GIWAXS studies were mainly focusing on the crystallization processes during spinning and annealing stages as well as different deposition techniques.

3.1. MA-Based Perovskites

3.1.1. MAPbI₃ Perovskites

Among versatile MA-based perovskites, MAPbI₃, with a bandgap of 1.53 eV, was the first to be considered as a suitable material for light absorption in perovskite photovoltaic devices, using a one-step spin-coating deposition method.^[32] To investigate the film formation mechanism, Amassian et al. monitored the crystallization pathway of MAPbI₃ during the one-step spin-coating process by in situ GIWAXS measurements.^[14b] The phase transition from a disordered precursor (a gel-like halo at low q values) to a highly ordered solvate phase was unveiled in Figure 6a. The film was then annealed and evolved to have a better surface coverage and device performance (Figure 6b). Nevertheless, the efficiency was only $\approx 10\%$ due to the dendritic

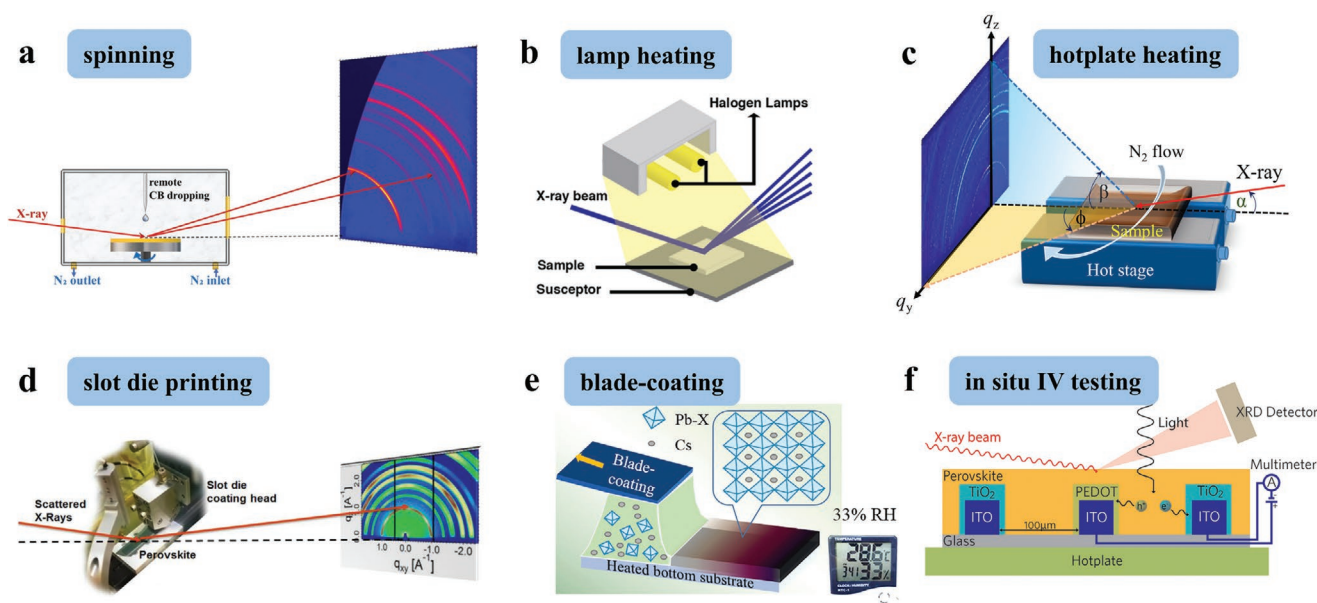


Figure 5. In situ GIWAXS measurements under various film deposition conditions, including: a) Spinning. Reproduced with permission.^[14a] Copyright 2019, Wiley-VCH. b) Lamp heating. Reproduced with permission.^[25] Copyright 2017, American Chemical Society. c) Hotplate heating. Reproduced with permission.^[26] Copyright 2017, American Chemical Society. d) Slot-die printing. Reproduced with permission.^[27] Copyright 2017, Springer Nature. e) Blade coating. Reproduced with permission.^[28] Copyright 2019, Elsevier. f) In operando IV testing. Reproduced with permission.^[29] Copyright 2018, Royal Society of Chemistry.

morphology with pinholes from the one-step spin-coating process (Figure 6c). Fortunately, incorporating additives in the perovskite precursor improves the film morphology by altering the crystallization process.^[33] For instance, the *N*-cyclohexyl-2-pyrrolidone (CHP) additive remarkably suppresses the formation of the solvate phase owing to the interaction between CHP molecules and PbI_2 , facilitating the growth of perovskite crystals based on in situ GIWAXS results in Figure 6d.^[30] Besides, CHP molecules can also serve as nucleation sites, thus forming a dense and compact perovskite film with small grains (Figure 6e). As a result, the best PCE of the device with CHP was $\approx 12.87\%$. In addition, in situ GIWAXS revealed that organic gelator additives such as 1,3:2,4-di-*O*-methylbenzylidene-*D*-sorbitol (MDBS) can delay the crystallization of the precursor solvate through the formation of a fibrillar 3D network, creating compact and highly crystalline structures.^[34] Antisolvent engineering is another strategy to improve the perovskite film morphology by quickly washing off the solvents and promoting crystallization.^[35] With the aid of in situ GIWAXS measurements in Figure 6f, it was found that the use of antisolvents in the one-step process suppresses the crystallization of PbI_2 , which is beneficial for the formation of a uniform perovskite film without pinholes.^[31] Furthermore, introducing small molecules, such as phenyl-C61-butyric acid methyl ester (PC₆₁BM) and 3,9-bis(2-methylene-(3-(1,1-dicyanomethylene) indanone))-5,5,11,11-tetrakis(4-hexylphenyl)-dithieno[2,3-d':2',3'-d']-s-indaceno[1,2-b:5,6-b']dithiophene (ITIC), into the antisolvent promote nucleation and the growth of the MAPbI_3 -DMSO solvate, serving as the basis for the further growth of MAPbI_3 perovskites.

Compared with the one-step spinning process, the two-step method involves sequential depositions of an inorganic PbI_2

layer followed by organic salts, thus enabling a more controllable crystallization process step by step.^[38] In the light of in situ GIWAXS results, it was found that the formation of PbI_2 in the first step experienced three stages, including the disordered solution (P_0) and sequentially formed PbI_2 -DMF crystalline phases (P_1 , P_2),^[36] as shown in Figure 7a. Among these phases, the solvated P_2 phase of PbI_2 -DMF can spontaneously convert to MAPbI_3 perovskites by the means of direct substitution or intercalation of DMF by MAI molecules due to the larger affinity between MA^+ and Pb^{2+} . The perovskite film fabricated from the P_2 phase exhibited smaller recombination losses and a smoother surface, thus delivering the best device performance with PCEs over 16% (Figure 7b). It suggests that the PbI_2 film deposited in the first step will influence the following perovskite crystallization process in the second step, which was further elucidated by Chikamatsu et al. who studied the real-time crystal transformation process with in situ GIWAXS by dropping MAI solutions onto PbI_2 films during the second step (Figure 7c).^[37] Along with the interdiffusion between PbI_2 and MAI, the perovskite peak gradually built up while concurrently the PbI_2 peak decreased, as shown in Figure 7d. Meanwhile, the corresponding polar intensity profiles in Figure 7e revealed a transition from preferred orientations to a random orientation along with the reaction time, suggesting that the perovskite crystals were initially oriented based on the crystal orientation of the PbI_2 phase (Figure 7f), but then gradually evolved into a random orientation.

After forming an as-cast film by a spin-coating process, thermal annealing treatments are generally required for the removal of solvents and the further growth of perovskite crystals.^[39] Therefore, the study on the perovskite crystallization process during annealing is equally important. As discussed

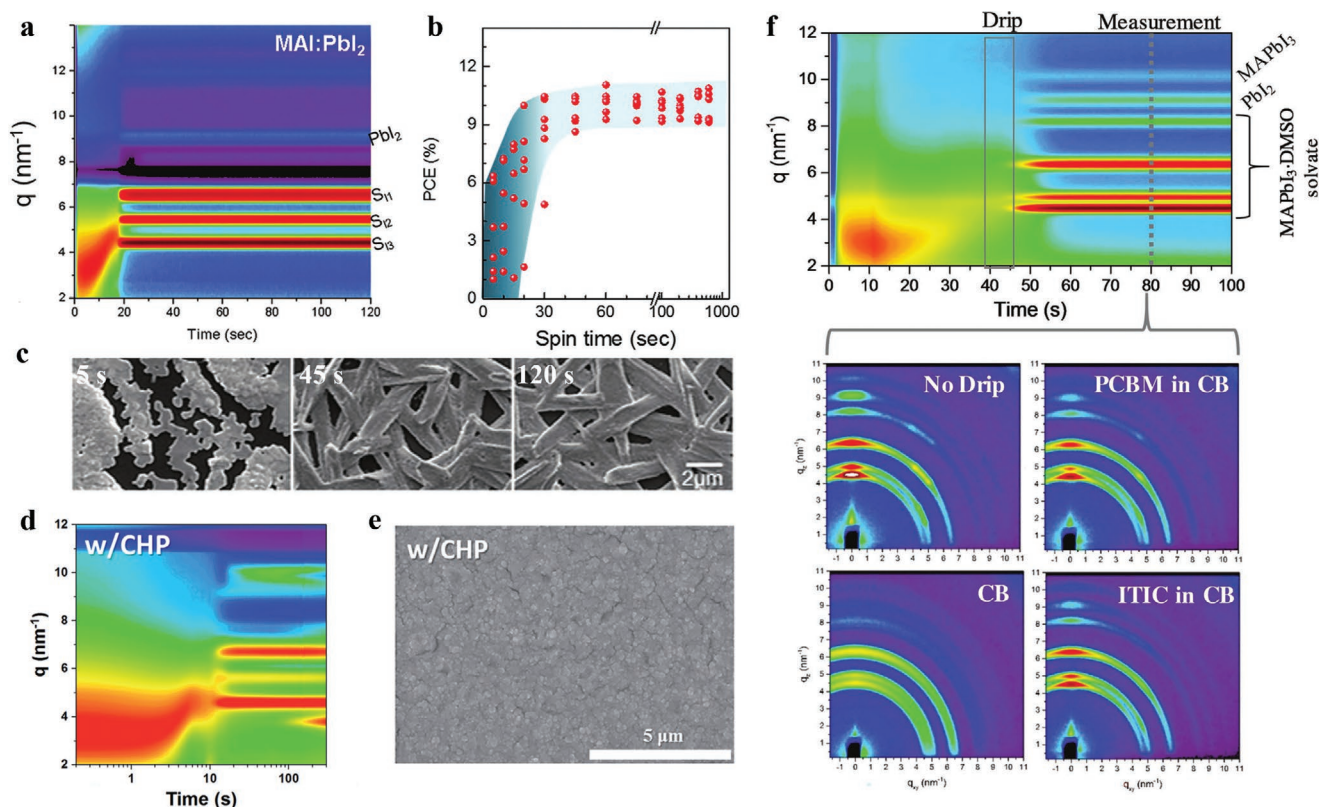


Figure 6. MAPbI₃ crystallization kinetics during the one-step spinning process. a) The false-color intensity map versus q and spinning time extracted from in situ GIWAXS data of the MAPbI₃ perovskite. The corresponding influence of the spinning time prior to thermal annealing on the b) device performance and c) film morphologies. Reproduced with permission.^[14b] Copyright 2016, Wiley-VCH. d) The in situ GIWAXS data and e) scanning electron microscopy (SEM) image of the MAPbI₃ perovskite with CHP additives. Reproduced with permission.^[30] Copyright 2020, Royal Society of Chemistry. f) In situ GIWAXS measurements performed at 80 s for the no-drip, CB drip, PCBM, and ITIC drips, and the corresponding GIWAXS patterns. Reproduced with permission.^[31] Copyright 2018, Wiley-VCH.

before, only an intermediate solvate phase can be crystallized in as-cast films after the one-step spin-coating process of MAPbI₃,^[14b] so Hest et al. monitored the succeeding phase transition from the intermediate phase to the MAPbI₃ perovskite phase during annealing using in situ GIWAXS.^[25] They also compared the annealing processes from different as-cast films with and without the antisolvent treatment and found that the role of the antisolvent is not to form an intermediate phase but to achieve a uniform MAPbI₃ film with minimal intermediate complexes. In other words, the intermediate compound cannot guarantee a high-quality pinhole-free perovskite film. However, Chen et al. proposed that DMSO-intermediate compounds can act as binders to merge adjacent perovskite grains and to eliminate grain boundaries, resulting in a perovskite film with overall larger grain sizes.^[40] They employed in situ GIWAXS characterization to demonstrate that, in comparison to DMF, the DMSO vapor can restrain the perovskite formation due to the higher coordinating capacity with perovskites (Figure 8a). Therefore, a dynamic equilibrium process between the formation and dissolution of perovskite crystals during solvent annealing can eventually reform the perovskite film morphology (Figure 8b).

The annealing processes of MAPbI₃ perovskite films fabricated through other techniques such as slot-die coating^[27] and

blade-coating^[41] methods have been also studied by in situ GIWAXS measurements, providing more detailed perovskite crystallization mechanisms. Gong et al. employed the slot-die coating method to deposit MAPbI₃ perovskite films and captured the time evolutions of perovskite crystallites formation under various temperatures, as shown in Figure 8c–f.^[27] At room temperature, an initial scattering halo was observed at $q \approx 0.4 \text{ \AA}^{-1}$ which was assigned to the [PbI₆]⁴⁻ octahedrons and the surrounding molecules. But it gradually shifted to a higher q value of $0.6 \text{ \AA}^{-1}$, indicating the continuous evaporation of DMF molecules. The solvent evaporation was slow at room temperature and the perovskite peak at $1.0 \text{ \AA}^{-1}$ was detected after dozens of minutes (Figure 8c). In contrast, higher annealing temperatures such as 60 and 80 °C not only significantly promoted the crystallization of perovskites within several seconds but also eliminated intermediate complexes, resulting in a direct formation of the perovskite phase from the precursors. However, prolonged heating at 100 °C gave rise to the decomposition of perovskite and the formation of PbI₂ (Figure 8f). The nano-assembly model of the perovskite formation from precursor to crystal is shown in Figure 8g. Similar phenomena were also observed in the blade-coating deposition process.^[42] A higher temperature led to the direct crystallization process from the disordered sol–gel precursor to the perovskite crystals

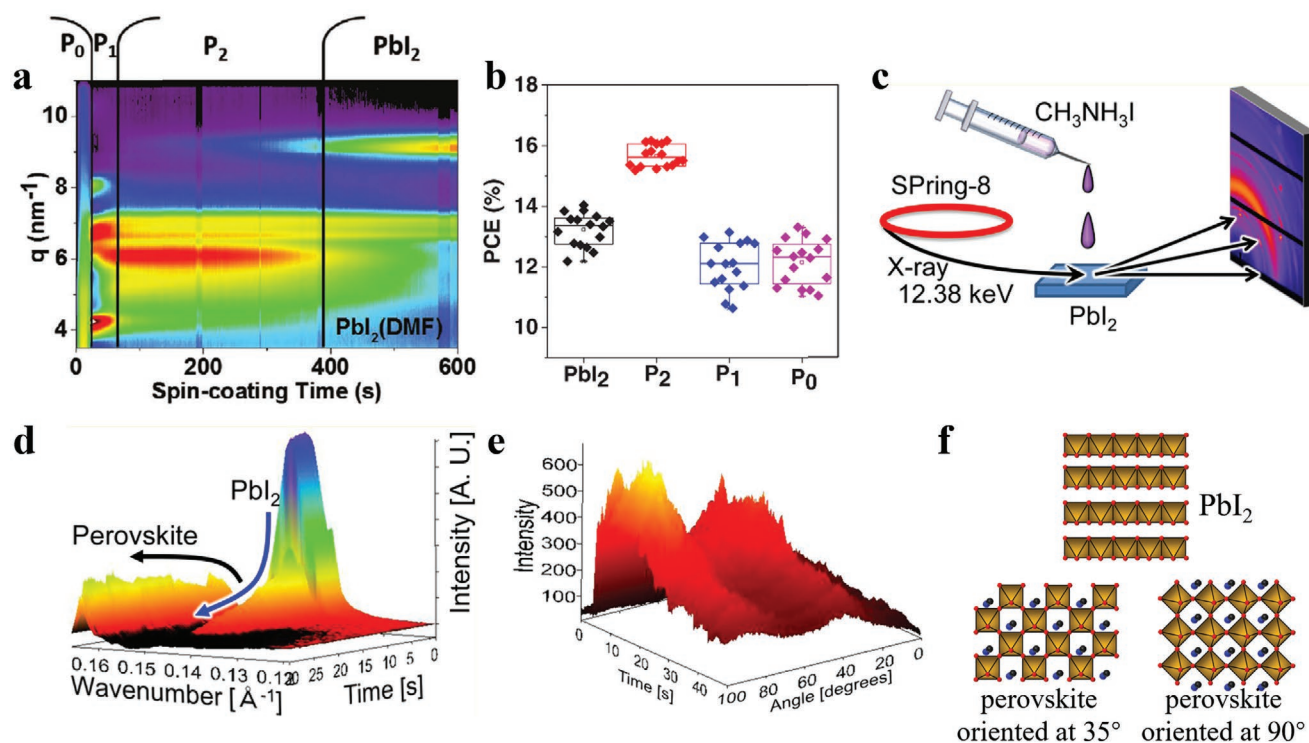


Figure 7. MAPbI₃ crystallization kinetics during the two-step spinning process. a) Time evolution of the GIWAXS intensity map of the PbI₂ film spinning process and b) the PCE distribution for PSCs fabricated from different PbI₂ phases. Reproduced with permission.^[36] Copyright 2019, Wiley-VCH. c) Schematic illustration of the in situ GIWAXS experimental setup employed to monitor the crystallization process during the two-step fabrication process. Time evolutions of d) peak intensity of PbI₂ and perovskite phases and e) the angle-dependent intensity for the film reacted at 70 °C. f) Crystal structures of PbI₂ and perovskites oriented at 35° and 90°. Reproduced with permission.^[37] Copyright 2015, American Chemical Society.

without intermediate complexes, which is important to form high-quality perovskite films and thus deliver a decent device efficiency.^[43] Apart from the optimized temperature, in situ GIWAXS can be also employed to determine the optimized annealing time during the blade-coating process based on the relative change of the perovskite peak area.^[44]

3.1.2. MAPbI_{3-x}Cl_x Perovskites

Since pure MAPbI₃ is vulnerable to moisture or heat attack and easily decomposes to PbI₂ and other complexes,^[46] smaller halide ions such as Br⁻ and Cl⁻ were introduced into MAPbI₃ to contract the lattice and stabilize the crystal structure.^[47] The incorporation of Br⁻ can greatly promote the formation of the perovskite phase, but on the other hand, enlarge the bandgap, sacrificing the light absorption.^[48] In comparison, Cl⁻ only serves as an intermediate element to assist the formation of perovskite films, and will escape from the film during thermal annealing.^[46] More interestingly, it was found that the isotropic polycrystalline MAPbI₃ perovskite film can be rearranged by Cl⁻ to form a film with a preferred orientation. By virtue of in situ GIWAXS measurements, McNeill et al. carefully investigated the microstructural evolution of MAPbI₃ films from randomly oriented grains to preferred OOP (110) oriented grains with MACl post-treatment.^[49] The chlorine-containing intermediate phase plays a crucial role as the mineral bridge to transform

primary, nanoscale perovskite grains into uniaxial oriented quasi-single crystal grains. Furthermore, Choi et al. reported that the perovskite orientation conversion by MACl propagated from top to bottom of the film,^[46] as shown in **Figure 9**. The preferred OOP (110) orientated grains are beneficial for charge transfer and collection and thus device performance.

In addition to the post-treatment with MACl, a more common method of incorporating Cl⁻ into MAPbI₃ is simply mixing the MAPbI₃ perovskite precursor with chlorine sources such as MACl and PbCl₂.^[52] The direct incorporation of Cl⁻ can also lead to preferential orientation of MAPbI_{3-x}Cl_x perovskites with the (110) plane parallel to the surface, and it persists throughout the whole annealing process,^[50] as reported by Lidzey et al. in **Figure 10a**. Furthermore, they employed in situ GIWAXS to establish the correlation between the integrated intensity of the perovskite peak ($x(t)$), annealing time, and device performance (Figure 10b). As a consequence, an optimized annealing time of 117 min was identified due to the highest value of $x(t)$ which indicates a complete conversion of intermediate phases to the perovskite phase without decomposition, thus leading to high efficiencies over 12%. As demonstrated by this work, the MAPbI_{3-x}Cl_x perovskite requires a longer annealing time and a higher annealing temperature to form a pure perovskite phase than the MAPbI₃ counterpart,^[46,50] because of the moderate sublimation of Cl⁻.^[53] To shorten the annealing time, Snaith et al. replaced lead halides (PbI₂ and PbCl₂) with non-halide lead acetate (PbAc₂), which

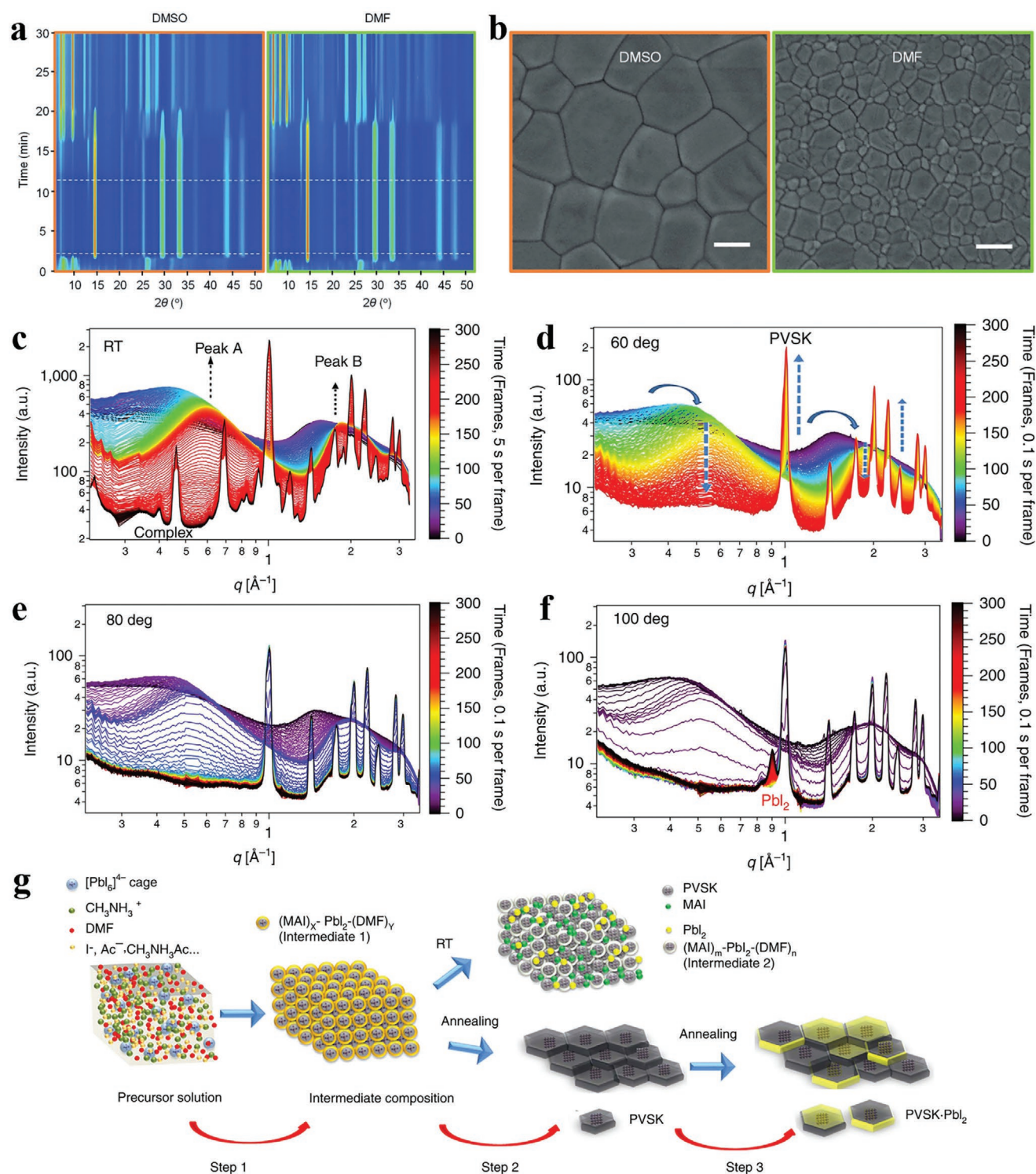


Figure 8. Annealing process of $MAPbI_3$. a) The in situ GIWAXS false-color intensity maps versus q and annealing time under DMF and DMSO vapors, and b) the corresponding SEM images of resulted perovskite films. Reproduced with permission.^[40] Copyright 2018, Wiley-VCH. c–f) In situ GIWAXS intensity profiles for the perovskite films annealed under different temperatures. g) The nano-assembly model of the perovskite formation from precursors to crystals. Reproduced with permission.^[27] Copyright 2017, Springer Nature.

allows a much faster perovskite crystal growth in a few minutes because of the facile removal of CH_3NH_3Ac in the light

of in situ GIWAXS results.^[54] An ultra-smooth and almost pin-hole-free perovskite film was obtained through the accelerated

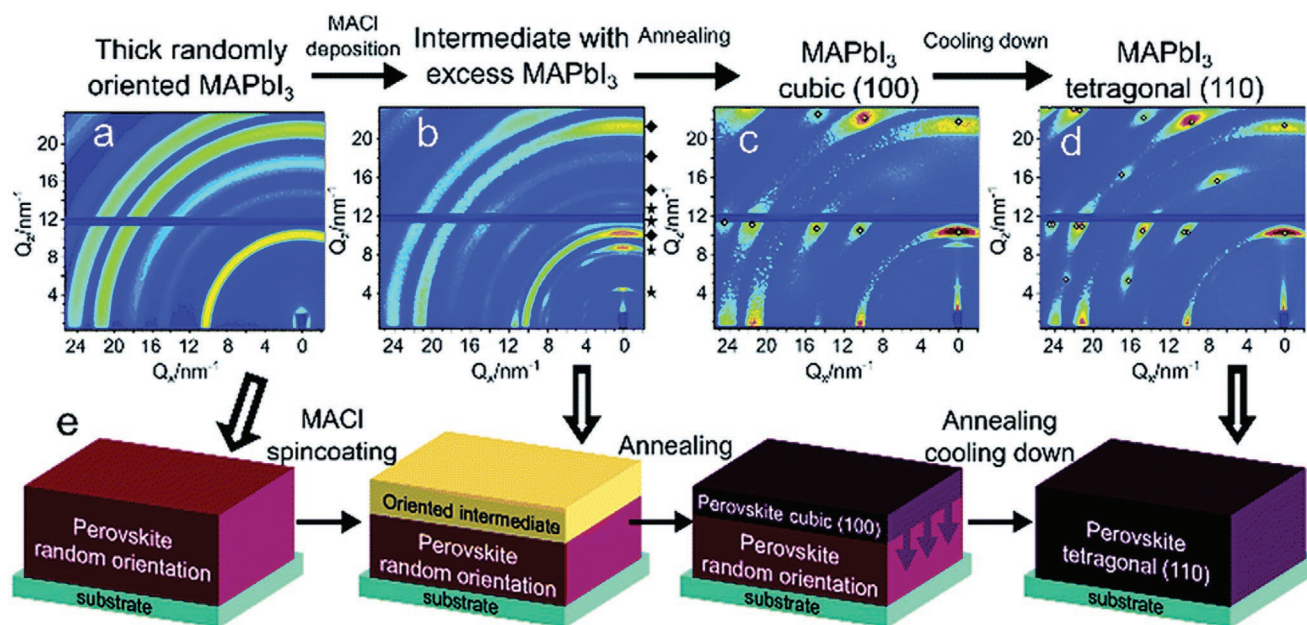


Figure 9. Structural evolution of MAPbI₃ perovskites with MACl post-treatment. a–d) Representative 2D GIWAXS patterns of MACl-treated perovskite films in different stages. e) Illustration of the proposed homoepitaxial mechanism. Reproduced with permission.^[45] Copyright 2017, Royal Society of Chemistry.

crystallization process. Estroff et al. further calculated the activation energy of perovskites with iodide, chloride, acetate, and nitrate ions from in situ GIWAXS data, demonstrating that the faster crystallization rate with acetate is ascribed to the lower activation energy.^[55] Besides, various additives such as polymers^[56] and PbS quantum dots^[57] were incorporated into the precursor solution, and in situ GIWAXS measurements revealed an acceleration in the MAPbI_{3-x}Cl_x perovskite crystal growth kinetics with these additives. As for the structural evolution of MAPbI_{3-x}Cl_x, Snaith and co-workers conducted in situ GIWAXS measurements and first identified three distinct structures during the annealing process of MAPbI_{3-x}Cl_x: a crystalline intermediate structure, a 3D perovskite structure, and a mixture of compounds as the products after degradations (Figure 10c).^[51] Later on, combining in situ GIWAXS, XRD, and X-ray photoemission spectroscopy (XPS), Jeng et al. identified this unresolved intermediate phase to be a 2D monoclinic perovskite MA₂Pb₂Cl₂MAI, which is composed of zigzagged lead-halide octahedra and sandwiched MAI layers (Figure 10d).^[26] The 2D structure of the intermediate phase is mainly aligned with the (110) planes of the 3D perovskite cubic phase, providing a template for the epitaxial growth of perovskites. Lidzey et al. further tracked the rotation and deformation of the intermediate and perovskite crystallite grains during annealing with in situ GIWAXS measurements, revealing a clockwise rotation of perovskite crystallites accompanied by a tiny lattice expansion.^[58]

Another piece of interesting work conducted by Lilliu et al. simultaneously monitored in situ photovoltaic and structural evolutions of MAPbI_{3-x}Cl_x-based PSCs during the perovskite film formation.^[29] They devised an interdigitated back-contact (IBC) device architecture to avoid the block of capping layers and directly exposed the perovskite film to light and X-ray

beams (Figure 11a). Current density–voltage (*J*–*V*) curves were correlated with the structural evolution of the film by in situ GIWAXS measurements. A high open-circuit voltage was observed at early stages of the precursor-to-perovskite conversion owing to the formation of a percolating pathway for charge transport in the intermediate phase, suggesting the remarkable defect tolerance of perovskite materials. In contrast, the photocurrent was progressively increasing with the growth of perovskite crystals to form continuous paths for charge collection.

Here, we summarize various phase transition processes of MA-based perovskites during different film deposition procedures obtained from in situ GIWAXS measurements and the corresponding synchrotron beamlines in Table 1. Generally, MA-based perovskites experience three crystallization stages: I) disordered precursor stage; II) intermediate and PbI₂ phases; III) perovskite phase. With some treatments such as additives or anti-solvent dripping, the formation of intermediate and PbI₂ phases in stage II can be suppressed. A high annealing temperature can even skip the crystallization stage II and lead to a direct phase transition from precursor-to-perovskite crystal. However, a prolonged annealing time can also cause the decomposition of perovskites. Furthermore, the incorporation of Cl[–] can promote the orientational order of perovskite crystals, resulting in better charge transport and device performance.

3.2. FA-Based Perovskites

With a comprehensive understanding of the film formation and degradation mechanisms of MA-based perovskites, scientists realized that MA-based perovskite films are intrinsically unstable under thermal stress.^[61] By contrast, FA-based perovskites such as FAPbI₃ exhibit a better heat resistance. A smaller

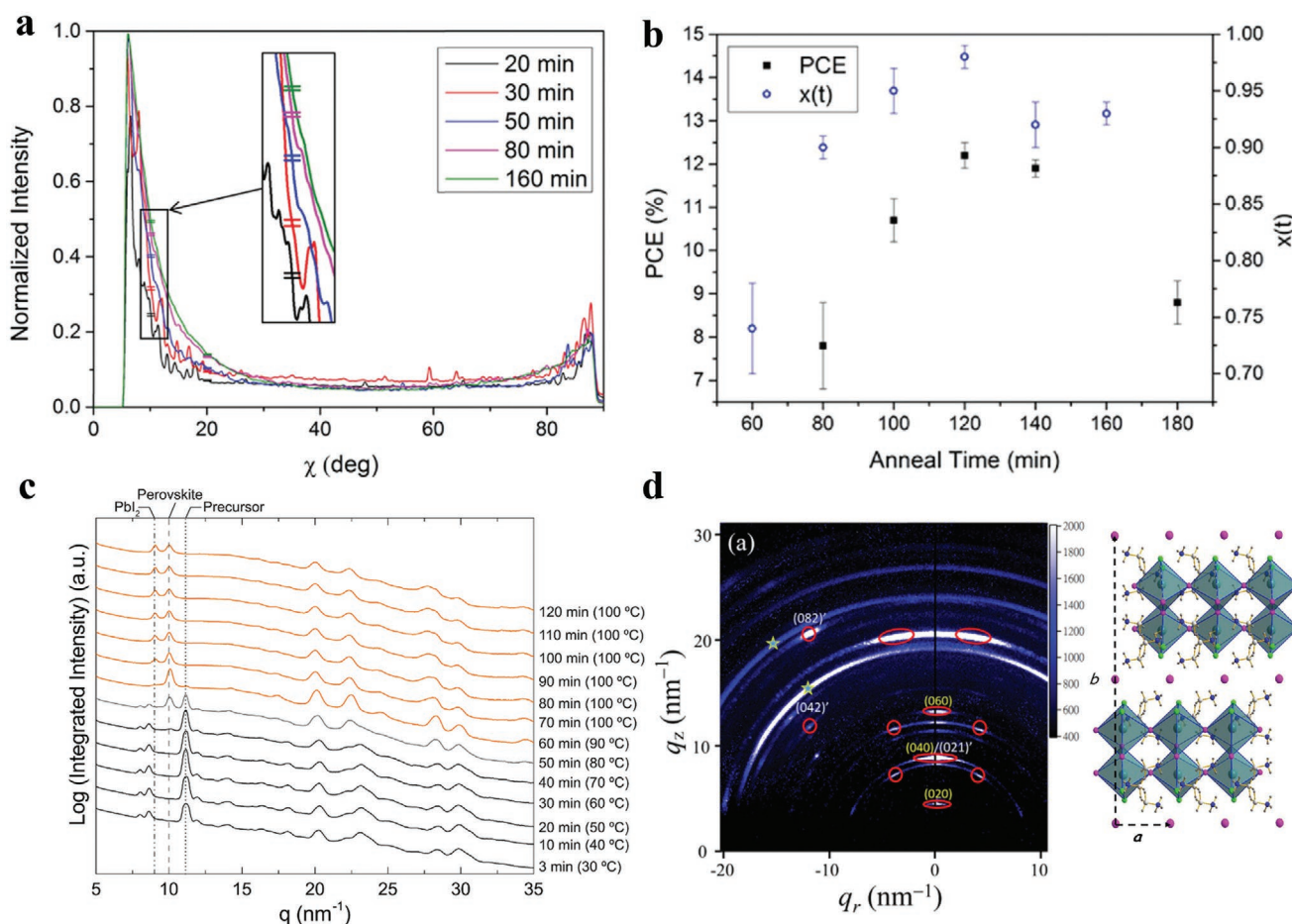


Figure 10. Annealing process of MAPbI_{3-x}Cl_x. a) Polar intensity profiles of the peak at $q \approx 1 \text{ \AA}^{-1}$ during the MAPbI_{3-x}Cl_x annealing at 80 °C extracted from GIWAXS results. b) PCE and $x(t)$ (defined as the normalized integrated intensity of the peak at $q \approx 1 \text{ \AA}^{-1}$ calculated via GIWAXS data) as a function of annealing time at 80 °C. Reproduced with permission.^[50] Copyright 2016, Wiley-VCH. c) Intensity profiles at key times during annealing of the MAPbI_{3-x}Cl_x film. Reproduced with permission.^[51] Copyright 2014, American Chemical Society. d) The GIWAXS pattern and crystalline structure of the intermediate phase. Reproduced with permission.^[26] Copyright 2017, American Chemical Society.

bandgap of FAPbI₃ is also beneficial for light absorption, contributing to the improved device efficiency.^[62] However, the photo-active cubic perovskite α -phase of FAPbI₃ is easily transformed into a photo-inactive hexagonal δ -phase,^[63] hindering the development of FA-based PSCs. The investigation on the crystallization and phase transitions during the formation of FAPbI₃ perovskites is therefore highly demanded. For one-step spin-coating fabricated pure FAPbI₃, in situ GIWAXS (Figure 12a,b) suggested that only the δ -phase is formed during the spinning process due to the low formation energy and thermal annealing is necessary to activate the phase transition from the δ -phase to the α -phase. This was further demonstrated by Amassian et al. that only the δ -phase or 2H phase was observed for FAPbI₃ perovskites during the spin-coating process,^[60] as shown in Figure 12c. In contrast, the incorporation of Br in FAPbI₃ will induce the formation of the 4H phase^[60] and promote the nucleation of the perovskite phase as well.^[64] It is more pronounced for pure FAPbBr₃ perovskites that the perovskite phase will be formed directly from a precursor state (Figure 12c).

As for the two-step spin-coating process, the same crystallization pathway of PbI₂ during the first step in Figure 7a was observed, and the metastable solvated polymorph P2 can easily react with FAI to form α -FAPbI₃ during the second step even without annealing (Figure 13).^[65] Nevertheless, further annealing is still required to further remove the residual solvent. A champion device with a PCE as high as 17.9% was achieved due to the reduced trap density and longer carrier lifetimes for the film formed from the P2 state. This work suggests that the state of the PbI₂ film in the first step is crucial for the resulted perovskite film quality. Huang et al., therefore, tried to alter the crystallization process of PbI₂ by employing ionic liquid methylamine formate (MAFa), leading to a uniform PbI₂ film with a preferential orientation.^[66] As a consequence, a stable α -FAPbI₃ perovskite film can be easily formed during the second step of dripping FAI, due to the improved diffusion path of FAI in the PbI₂ thin films and corner-sharing layered structure of PbI₂.

The spin-coating process can deposit an as-cast FAPbI₃ film, while thermal annealing is still a prerequisite to deliver

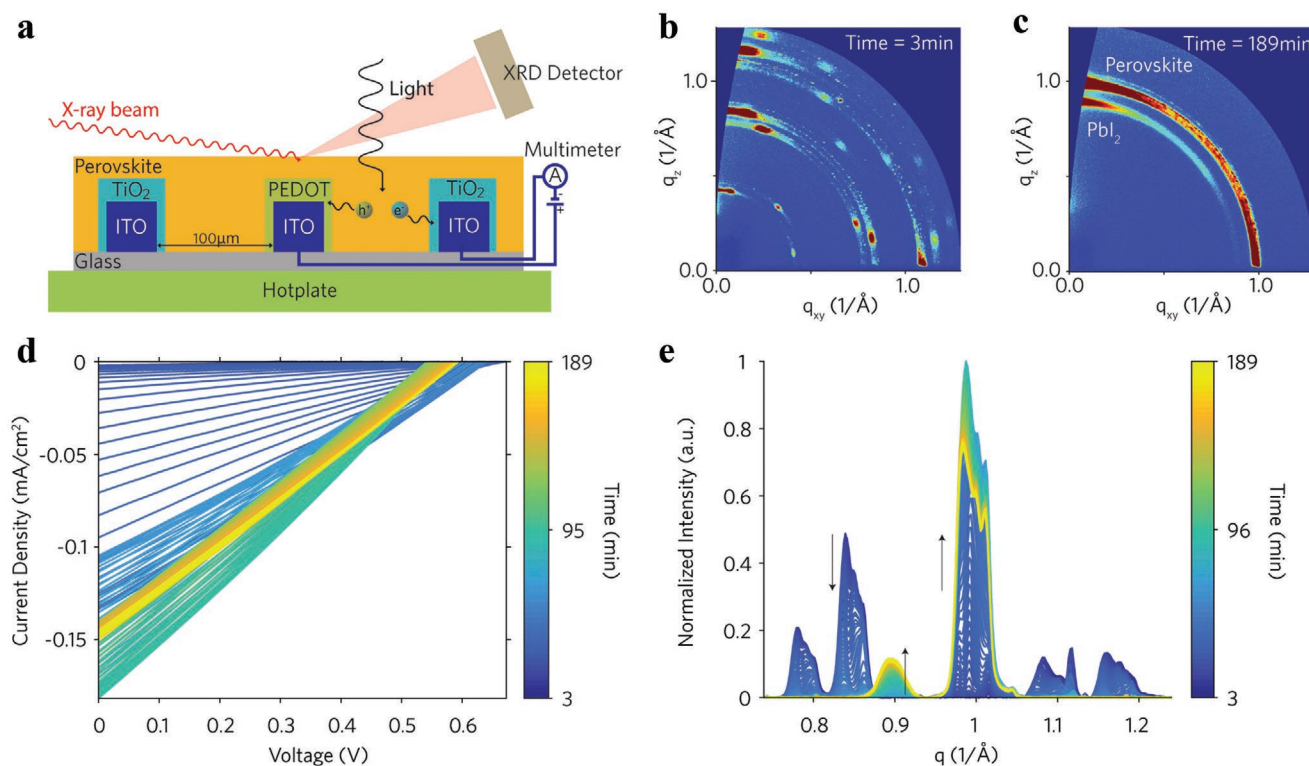


Figure 11. Simultaneous in situ GIWAXS and *I*-*V* measurements for a MAPbI_{3-x}Cl_x-based PSC. a) The device structure of an interdigitated back-contact (IBC) solar cell. 2D GIWAXS patterns of the film after b) ≈3 min and c) ≈189 min from the beginning of the anneal. d) *J*-*V* curves of IBC solar cells measured under light and e) the corresponding GIWAXS intensity profiles during the anneal. Reproduced with permission.^[29] Copyright 2018, Royal Society of Chemistry.

Table 1. Phase transitions of MA-based perovskites during different film deposition procedures obtained from in situ GIWAXS measurements and the corresponding synchrotron beamlines.

Perovskite	Deposition procedure	Phase transition	Treatment or remark	Beamline	Ref.
MAPbI ₃	One-step spinning	Precursor state → intermediate and PbI ₂ phases	CHP/MDBS: suppress the formation of the intermediate phase	D1, CHESS	[14b,30,34]
	Two-step spinning	Step 1: P0→P1→P2→PbI ₂ ;	Anti-solvent: suppress the formation of PbI ₂	D1, CHESS	[31]
		Step 2: PbI ₂ →perovskite	P2 is the best state to form perovskite crystals	D1, CHESS	[36]
	Annealing	Intermediate phase → perovskite	Perovskite orientation is initially based on the PbI ₂ crystals	BL46XU, JASRI	[37]
	Slot-die coating/blade coating	Room temperature: precursor → intermediate phase → perovskite/PbI ₂ Moderate temperature (≈60–80 °C): precursor → perovskite High temperature (≥100 °C): precursor → perovskite → perovskite/PbI ₂	DMSO solvent annealing can merge grains forming larger perovskite grains	BL14B1, SSRF	[40]
MAPbIBr ₂	One-step spinning	Precursor → intermediate phase/perovskite		BL7.3.3, ALS	[27]
MAPbI _{3-x} Cl _x	One-step spinning	Orientation from random to (110) preferred orientation	Post-treatment of MAPbI ₃ with MACl		
	Annealing	Intermediate phase → perovskite → perovskite/PbI ₂	Orientation start from the beginning	D1, CHESS	[14b,59]
			PbAc ₂ /PbS/polymers can accelerate crystallization	BLI22, Diamond	[45]
				BL23A, NSRRC	[50–51]
					[54,56–57]

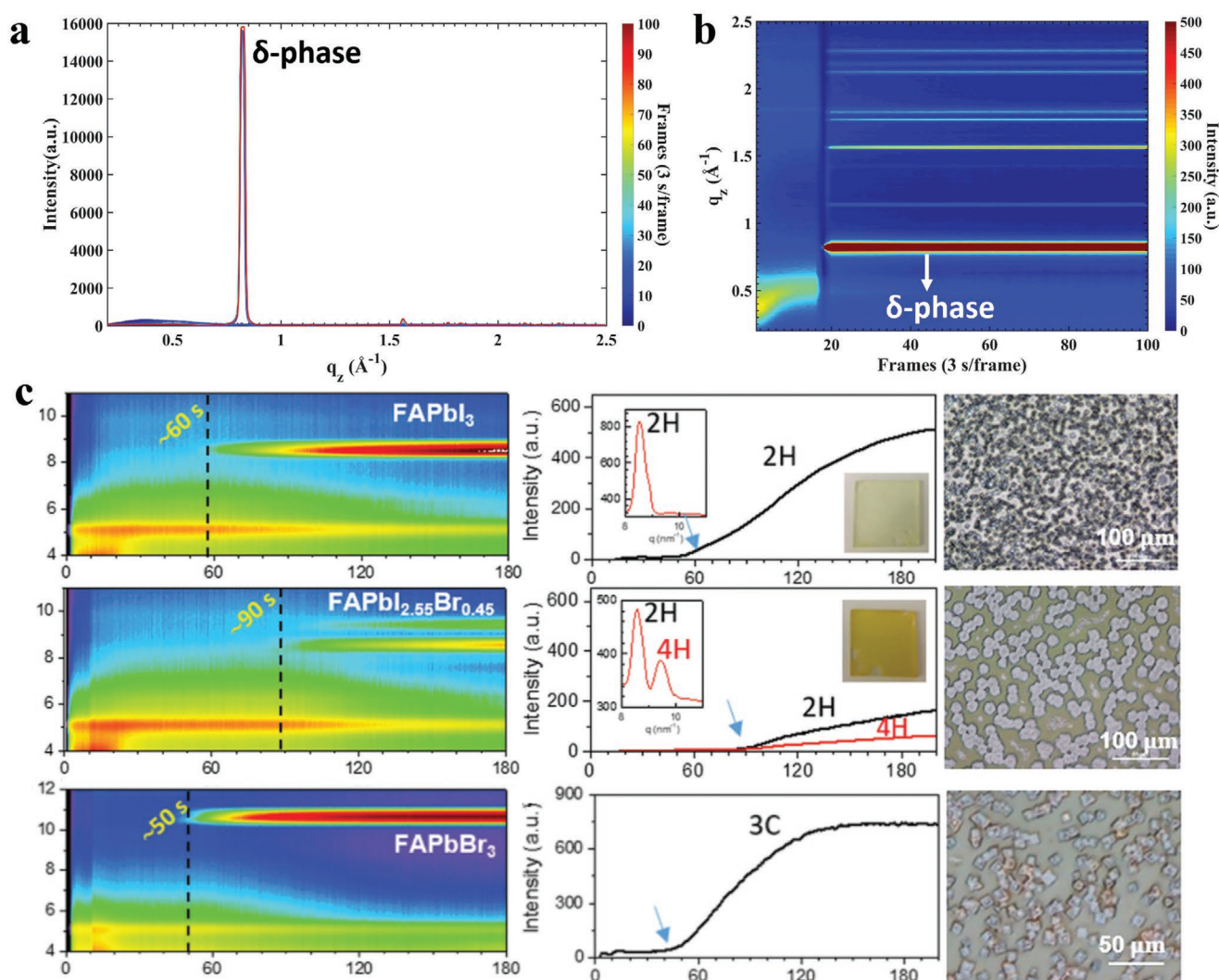


Figure 12. Crystallization of FAPbI₃ perovskites during the one-step spin-coating process. a) Intensity profiles and b) the false-color intensity map based on in situ GIWAXS data of the FAPbI₃ film during the one-step spinning (unpublished). c) Time-resolved GIWAXS intensity maps of FAPbI₃, FAPbI_{2.55}Br_{0.45}, and FAPbBr₃ perovskites during spin coating, and the corresponding optical microscopy images of the films. Reproduced with permission.^[60] Copyright 2019, Wiley-VCH.

high-quality FAPbI₃ films. Toney et al. systematically studied the temperature-induced crystal structural transformation and degradation of FAPbI₃ during annealing,^[67] as shown in Figure 14a. The film started in the δ -FAPbI₃ phase (named as precursor in Figure 14a), which slowly transitioned to the α -FAPbI₃ phase at the transition temperature of ≈ 130 °C. Elevated temperatures (170 and 330 °C) accelerated the complete phase transition in dozens of seconds, yet the PbI₂ phase was produced from the perovskite degradation with a longer annealing time. By coupling the in situ GIWAXS data with device performance, they successfully mapped the processing phase space of FAPbI₃ and corresponding device efficiencies as a function of annealing temperature and time in Figure 14b. An optimized processing condition, that is, annealing at 170 °C for 40 s, was found to deliver the best device efficiency with a pure α -FAPbI₃ phase. Apart from the annealing temperature, additive ions can alter the crystallization rate of FAPbI₃ perovskites as well.

Phenethylammonium (PEA⁺) and 4-fluoro phenethylammonium (FPEA⁺) cations were introduced into FAPbI₃ to retard the phase conversion kinetics by inducing a strain at the hetero-interface, as demonstrated by in situ GIWAXS data, thus enabling the growth of tenfold enlarged perovskite crystals.^[69] Chen et al. also employed in situ GIWAXS technique to investigate the influence of MA⁺, Br⁻, and Cl⁻ ions in pure FAPbI₃.^[68] An accelerated perovskite crystal growth speed in the order of MA⁺ > Br⁻ > Cl⁻ > pristine was observed (Figure 14c), accompanied with increasing perovskite crystallinity. The orientation of FAPbI₃ perovskites can be also regulated by the mixed ions from two preferential orientations to one dominant orientation with (111) planes parallel to the substrate (Figure 14d,e). In fact, the FAPbI₃ perovskite with mixed cations (MA⁺, FA⁺) and mixed halides (I⁻, Br⁻) is classified as the mixed perovskite, which is the most common perovskite material in high-efficiency PSCs. A further discussion on mixed perovskites comes in the following section.

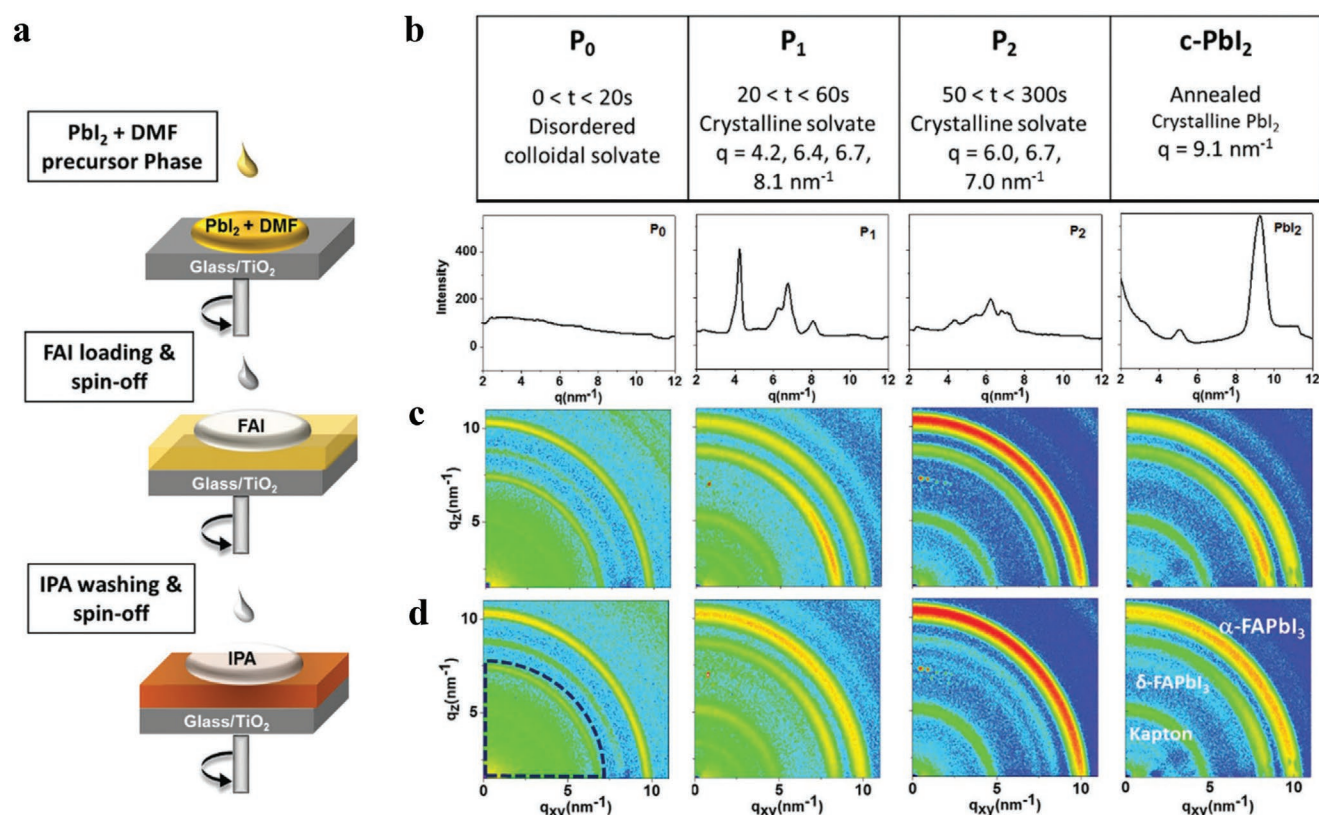


Figure 13. Crystallization of FAPbI₃ perovskites during the two-step spin-coating process. a) Schematics of the film deposition process employed during in situ GIWAXS measurements. b) The four stages of PbI₂ and c) the corresponding GIWAXS patterns. d) Representative GIWAXS patterns of FAPbI₃ films formed from different PbI₂ states. Reproduced with permission.^[65] Copyright 2020, Wiley-VCH.

3.3. Mixed Perovskites

3.3.1. MA-FA Mixed Perovskites

For the purpose of stabilizing the α -phase and suppressing the δ -phase of FAPbI₃, smaller cations (i.e., MA⁺, Cs⁺) and anions (i.e., Br) are usually introduced into the FAPbI₃ crystal lattice simultaneously to form the aforementioned mixed perovskites. The PSCs based on the benchmark mixed perovskites with a composition of MA_{0.17}FA_{0.83}Pb(I_{0.83}Br_{0.17})₃ can readily deliver high efficiencies exceeding 19% owing to the excellent photoelectrical properties.^[11,70] However, the complex composition of mixed perovskites inevitably complicates the phase evolutions during the film formation process, challenging the fundamental understanding of the crystallization mechanism.

To address this issue, the detailed crystallization process of mixed perovskite MA_{0.17}FA_{0.83}Pb(I_{0.83}Br_{0.17})₃ during one-step antisolvent spin coating has been investigated systematically by in situ GIWAXS characterization.^[14a] As shown in **Figure 15a**, three crystallization stages have been successfully identified, including the precursor state (stage I), the hexagonal δ -phase or 2H phase (stage II), and complex phases (stage III) including the intermediate phase, hexagonal 4H, 6H polycrystals, and the perovskite α -phase. The incorporation of Cs⁺ was found to suppress the formation of 4H, 6H as well as the intermediate phases formed in stage III, and the perovskite with Cs⁺ only

experiences the first two stages with the perovskite α -phase appeared as early as stage II (**Figure 15b**). By coupling with the device performance (**Figure 15c**) and ex situ characterizations, the concept of “annealing window” was proposed, which refers to the optimal timing to anneal the as-cast film after spin coating to avoid the formation of unwanted secondary polymorphs. The end of the “annealing window” is determined by the appearance time of the complex phases including the intermediate phase and 4H, 6H phases, which is revealed by in situ GIWAXS results in **Figure 15a**. The role of Cs⁺ in the MA-FA mixed perovskites was further confirmed by Amassian et al. in another work.^[71] They also explored the influence of Rb⁺ in the perovskite crystallization dynamics and found that the addition of Cs⁺ and Rb⁺ drastically suppressed the phase segregation into MA-I- and FA-Br-rich phases and promoted the conversion of the α -phase. In addition, the incorporation of Rb⁺ and Br⁻ was reported to widen the antisolvent processing window during the one-step spin-coating fabrication of high-quality perovskite films and efficient devices,^[60] as shown in **Figure 15d**. The antisolvent processing window widened from ≈ 60 to 200 s along with the doping of Br⁻ and Rb⁺ during spinning, thus improving the reproducibility of efficient PSCs.

In contrast to the one-step spin-coating process in which exists “annealing window” or “antisolvent processing window” that should be captured timely for the fabrication of high-quality perovskite films, the two-step fabrication procedure

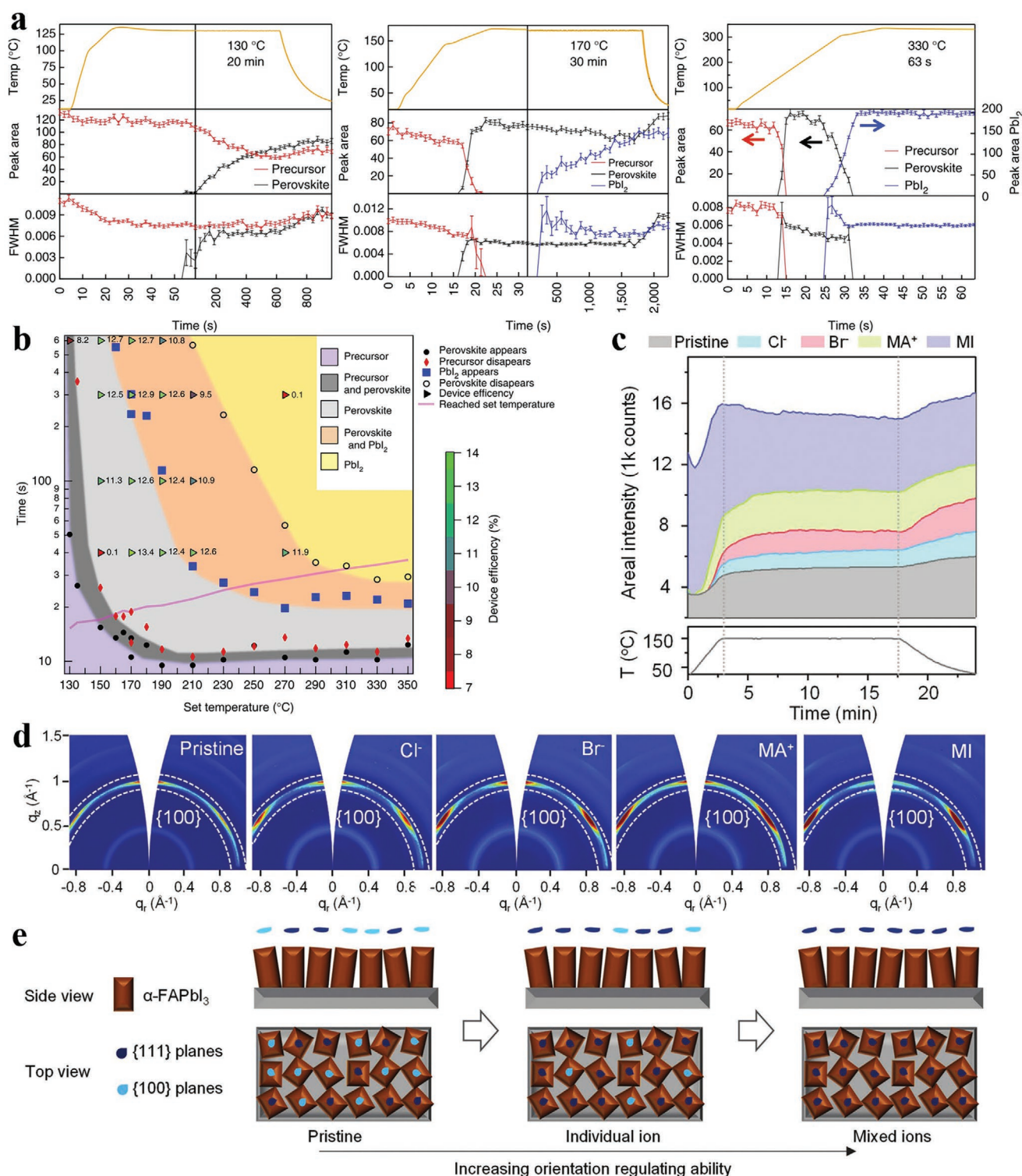


Figure 14. Crystallization analysis of FA-based perovskites during annealing. a) Time evolutions of integrated peak intensities for the δ -FAPbI₃ (denoted as precursor), α -FAPbI₃, and PbI₂ phases for films annealed at different temperatures. b) Processing phase space determined from in situ GIWAXS results and relevant device efficiencies. Reproduced with permission.^[67] Copyright 2017, Springer Nature. c) The peak area evolutions of the α -FAPbI₃ characteristic peak for the films doped with three different ions (Cl⁻, Br⁻, and MA⁺) during the annealing process. The undoped film and the film doped with all three ions are denoted as pristine and MI, respectively. d) The corresponding 2D GIWAXS patterns of the different perovskite films. e) Schematic illustration of the perovskite orientations with different ion recipes. Reproduced with permission.^[68] Copyright 2019, Wiley-VCH.

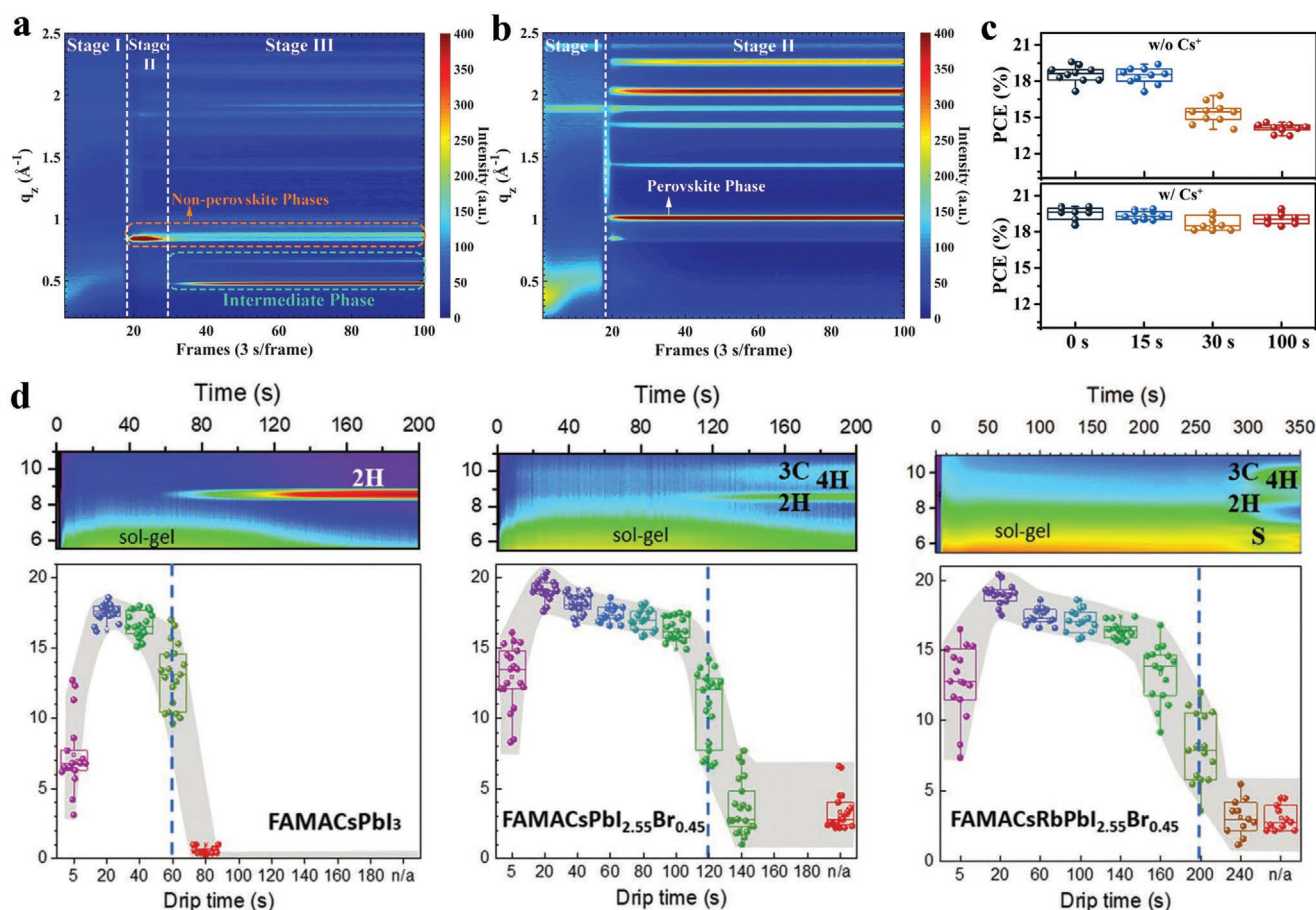


Figure 15. Mixed-perovskite crystallization during one-step spinning. The false-color intensity maps versus q_z and frame numbers extracted from in situ GIWAXS results measured during the one-step spin-coating process of the $\text{FA}_{0.83}\text{MA}_{0.17}\text{Pb}(\text{I}_{0.83}\text{Br}_{0.17})_3$ perovskite films a) without and b) with 10% Cs^+ . c) Statistical device performance of PSCs based on the corresponding perovskite films annealed with different spinning time after the chlorobenzene dripping. Reproduced with permission.^[14a] Copyright 2019, Wiley-VCH. d) In situ 2D GIWAXS intensity maps and the corresponding device performance fabricated with varied antisolvent dripping time for FAMACsPbI_3 , $\text{FAMACsPbI}_{2.55}\text{Br}_{0.45}$, and $\text{FAMACsRbPbI}_{2.55}\text{Br}_{0.45}$ perovskites. Reproduced with permission.^[60] Copyright 2019, Wiley-VCH.

enables a precise control of the perovskite film formation by decoupling the PbI_2 and perovskite crystallization processes and the roles of Cs^+ can be different as well.^[72] As thus, in situ GIWAXS was employed to monitor the perovskite crystallization kinetics of mixed perovskites with sequential doping of Cs^+ and GA^+ during the two-step spinning process (Figure 16a).^[15] The incorporation of Cs^+ in the first step induces the formation of the orthorhombic $\delta\text{-CsPbI}_3$ phase, offering an additional phase transition pathway from $\delta\text{-CsPbI}_3$ to perovskite besides the PbI_2 -to-perovskite pathway. However, sparse $\delta\text{-CsPbI}_3$ crystals cause the nonuniform nucleation and hence pinholes in the film, as shown in Figure 16b. Fortunately, incorporating GA^+ in the second step not only accelerates the phase transition from $\delta\text{-CsPbI}_3$ to perovskite (Figure 16c), but also greatly eliminates pinholes and improves the crystal grain size, thus pushing the efficiency of PSCs over 23%.

3.3.2. Cs-FA Mixed Perovskites

Despite the high efficiency of PSCs based on MA-FA mixed perovskites, the poor thermal stability of the MA-FA mixed

perovskites has been an issue due to the relatively volatile MA cations.^[73] Therefore, MA-free mixed perovskites (Cs,FA) $\text{Pb}(\text{I,Br})_3$ were then widely explored. Besides, they usually possess a wider bandgap, offering possible applications in tandem solar cells. The structural evolution of the mixed perovskite $\text{Cs}_{0.17}\text{FA}_{0.83}\text{Pb}(\text{Br}_{0.17}\text{I}_{0.83})_3$ during one-step spinning was investigated by Nogueira et al. through in situ GIWAXS measurements.^[74] They found that the perovskite crystallization pathway is influenced by the ambient humidity and precursor solvent ratio, as shown in Figure 17. On the one hand, the perovskite phase solely formed at a low humidity level (3–5% rH) while the increased humidity level led to the formation of unwanted hexagonal 2H, 4H, and intermediate phases, which suggests the mixed perovskites are sensitive and vulnerable to a humid environment. On the other hand, a higher amount of DMSO in the perovskite precursor prolonged the duration of the colloidal state, extending the time window for antisolvent injection in favor of forming a uniform perovskite film.

Despite the different crystallization kinetics of mixed perovskites during spinning, their behavior is similar to MAPbI_3 perovskites (Figure 8e) under an elevated processing temperature. The Cs-FA mixed perovskites also experience a direct

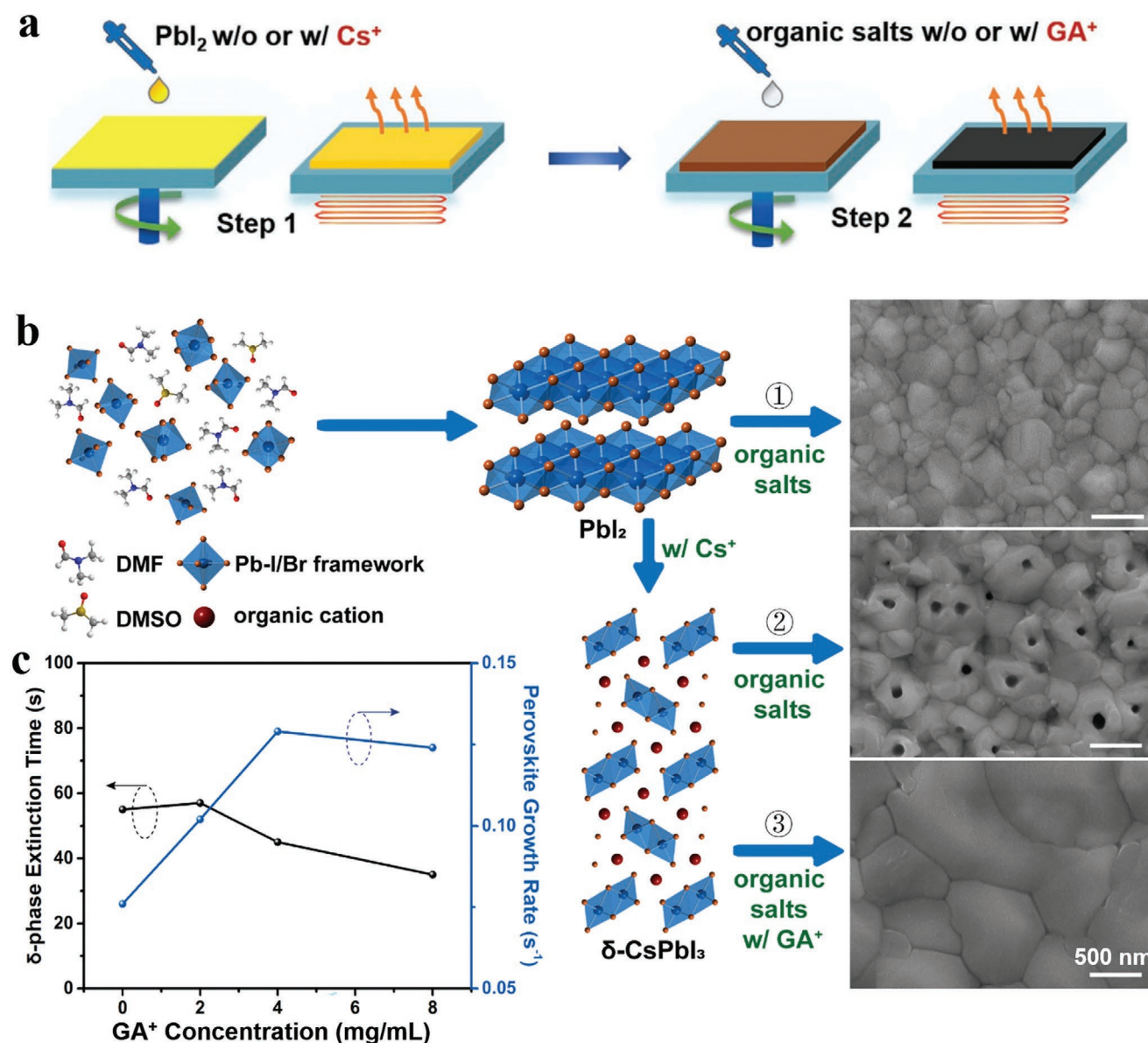


Figure 16. Mixed-perovskite crystallization during two-step spinning. a) Schematics of the two-step fabrication procedure of the perovskite film doped with Cs^+ and GA^+ during the first and second steps, respectively. b) Three phase transition pathways for the films without doping, with Cs^+ , with Cs^+ & GA^+ , and the corresponding SEM images. c) δ - CsPbI_3 phase extinction time and perovskite growth rate with different concentrations of GA^+ extracted from in situ GIWAXS results. Reproduced with permission.^[15] Copyright 2020, Wiley-VCH.

phase transition from the disordered precursor to the perovskite phase during the blade-coating process.^[64,75]

To compare the crystallization kinetics of FAPbI_3 and mixed perovskites, the corresponding phase transitions under various film deposition procedures are summarized in Table 2. In short, the formation of pure FAPbI_3 perovskites during one-step spinning usually experiences the phase transition from the precursor to the hexagonal δ -phase to the perovskite phase, whereas a two-step spinning method offers a distinct phase transition from PbI_2 to FAPbI_3 at room temperature. It is noted that a much higher annealing temperature can still engender the decomposition of FAPbI_3 to the PbI_2 phase. In comparison to the pure FAPbI_3 , mixed perovskites ((FA,MA)

Pb(I,Br)_3) exhibit complicated crystallization processes, typically from the precursor to the 2H phase to complex phases including the intermediate phase, hexagonal 4H, 6H polycrystals, and the perovskite α -phase. This complex crystallization process may be responsible for the poor reproducibility of mixed perovskite PSCs. Fortunately, incorporating A-site cations, such as Cs^+ and Rb^+ , can largely simplify the phase transition process by suppressing the formation of intermediate phases and secondary hexagonal phases, thus improving the device performance as well as reproducibility. Interestingly, doping Cs^+ in the two-step spinning process plays a disparate role in the crystallization kinetics, which induces the formation of the δ - CsPbI_3 phase in the first step, imposing a new phase transition pathway

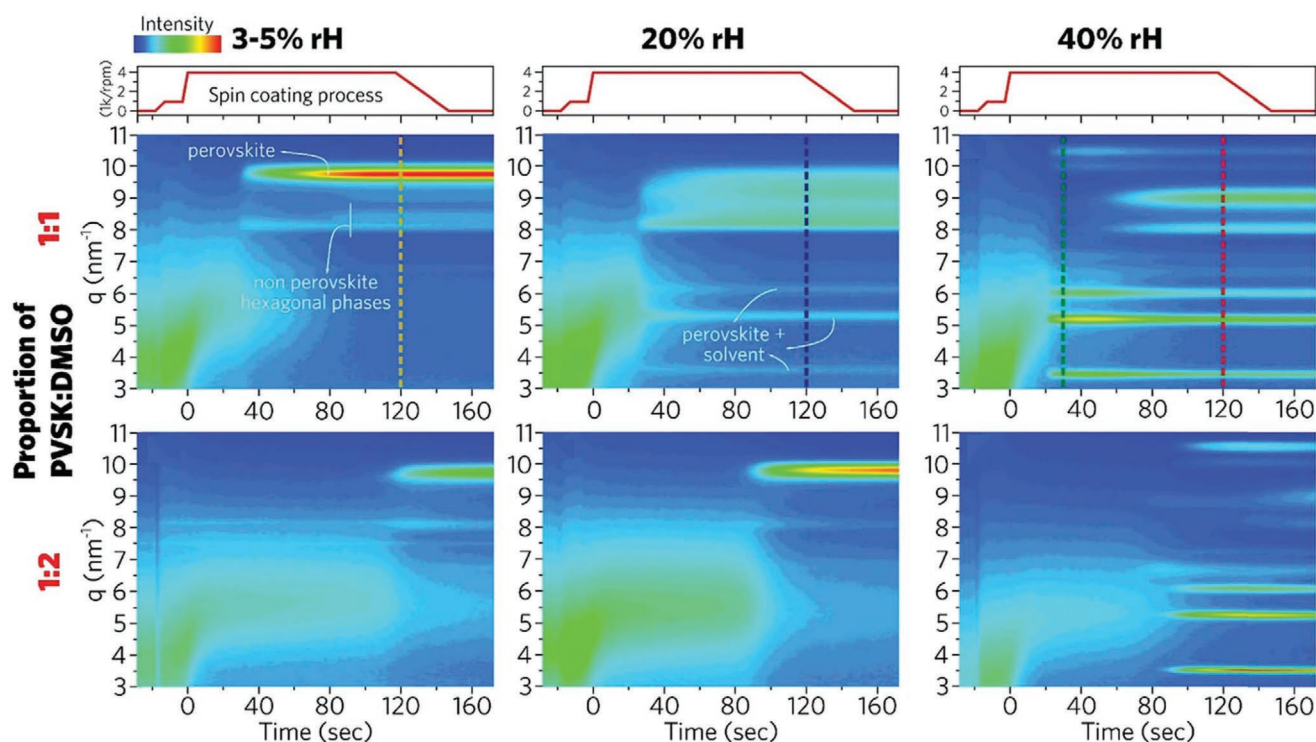


Figure 17. Crystallization processes of the $\text{Cs}_{0.17}\text{FA}_{0.83}\text{Pb}(\text{Br}_{0.17}\text{I}_{0.83})_3$ perovskites under different relative humidity (rH) and perovskite:DMSO ratios. Reproduced with permission.^[74] Copyright 2019, Royal Society of Chemistry.

from the sparse $\delta\text{-CsPbI}_3$ nucleation sites to perovskite phase in addition to the original PbI_2 to perovskite pathway. And the crystallization rate of the new pathway can be promoted by GA^+ doping in the second step, giving rise to a uniform perovskite

film with a large grain size. In addition, the film deposition environment such as humidity and temperature can alter the crystallization kinetics, and the delicate control of the environment is critical for the formation of high-quality perovskite films.

Table 2. Phase transitions of pure FAPbI_3 and mixed perovskites during different film deposition procedures obtained from in situ GIWAXS measurements and the corresponding synchrotron beamlines.

Perovskite	Deposition procedure	Phase transition	Treatment or remark	Beamline	Ref.
FAPbI_3	One-step spinning	Precursor $\rightarrow$ hexagonal δ -phase		BL23A, NSRRC	[60]
	Two-step spinning	Step 1: $\text{P0} \rightarrow \text{P1} \rightarrow \text{P2} \rightarrow \text{PbI}_2$; Step 2: $\text{PbI}_2 \rightarrow$ perovskite.	P2 state is the best polymorph to form the perovskite.	D1, CHESS	[65]
	Annealing	Low temperature ($\leq 130^\circ\text{C}$): hexagonal δ -phase $\rightarrow$ perovskite; High temperature ($\geq 170^\circ\text{C}$): hexagonal δ -phase $\rightarrow$ perovskite $\rightarrow$ perovskite/ $\text{PbI}_2 \rightarrow \text{PbI}_2$		7-2, SSRL	[67]
(FA, MA) $\text{Pb}(\text{I}, \text{Br})_3$	One-step spinning	Pristine: precursor $\rightarrow 2\text{H} \rightarrow 4\text{H}/6\text{H}/\text{intermediate phase}/\text{perovskite}$ Cs^+ doping: precursor $\rightarrow 2\text{H}/\text{perovskite}$	The perovskite film should be annealed within the annealing window to achieve high efficiencies of PSCs.	BL23A, NSRRC	[14a]
		Br^- & Rb^+ doping: precursor $\rightarrow 2\text{H}/4\text{H}/\text{perovskite}$	The processing window of anti-solvent is widened by Br^+ and Rb^+ doping.	D1, CHESS	[60]
	Two-step spinning	Pristine: precursor $\rightarrow \text{PbI}_2 \rightarrow \text{intermediate phase}/\text{perovskite}$ Cs^+ doping: precursor $\rightarrow \delta\text{-CsPbI}_3/\text{PbI}_2 \rightarrow \text{perovskite}$	GA^+ cations accelerate the phase transition from $\delta\text{-CsPbI}_3$ to perovskite.	BL23A, NSRRC	[15]
(FA, Cs) $\text{Pb}(\text{I}, \text{Br})_3$	One-step spinning	Low rH: precursor $\rightarrow$ hexagonal δ -phase/ perovskite High rH: precursor $\rightarrow 2\text{H}/4\text{H}/\text{intermediate phase}$	A higher DMSO ratio extends the time window of anti-solvent injection.	XRD2, LNLS	[74]
	Blade coating	Low temperature: precursor $\rightarrow$ hexagonal δ -phase/perovskite High temperature: precursor $\rightarrow$ perovskite		7-2/11-3, SSRL	[75]

3.4. Cs-Based All-Inorganic Perovskites

The thermal stability issue of PSCs has been an obstacle for commercialization because of the relatively high volatility of the commonly employed organic cations like MA⁺ and FA⁺. The replacement of organic cations with inorganic cations, such as Cs⁺, forming all-inorganic perovskites, is an effective means to enhance the thermal stability of PSCs.^[76] The first piece of work on all-inorganic CsPbI₃ PSCs with an efficiency of 2.9% was reported in 2015,^[77] demonstrating a feasible and promising research direction toward thermally stable all-inorganic PSCs. However, since Cs⁺ is too small as an A-site cation to maintain a stable cubic structure, CsPbI₃ perovskites tend to mutate from the photo-active black phase (i.e., α , β , γ -CsPbI₃) to the photo-inactive yellow phase (δ -CsPbI₃) at room temperature. To suppress the unwanted spontaneous phase transition, many efforts have been devoted to stabilizing the black phase such as the optimization of deposition procedures,^[78] surface passivation,^[79] interfacial engineering,^[80] and compositional engineering,^[81] pushing the efficiency of all-inorganic PSCs over 20%.^[79] Particularly, compositional X-site engineering by introducing smaller halide ions Br[−] or Cl[−] is a widely used approach for the purpose of adjusting the tolerance factor and thus achieving a stable cubic phase.^[82] In the meantime, the incorporation of Br[−] or Cl[−] ions in CsPbI₃ can largely affect the film formation process and the corresponding device performance. Therefore, investigations on the crystallization mechanism of CsPbX₃ (X = I, Br, or Cl) are particularly important on the road of efficiency improvement.

The influence of ternary halide mixing on the phase evolution of CsPbX₃ during the spinning process was systematically studied in virtue of synchrotron-based in situ GIWAXS characterizations.^[83] Br[−] ion was found to play an important role

in facilitating the formation of the black phase as well as suppressing the unwanted phase transition to the yellow phase, as shown in **Figure 18**. It was ascribed to the higher electronegativity and smaller ionic size of Br[−] than those of I[−], resulting in stronger coordination of Cs⁺ ions and reduced lattice distortion. In addition, Cl[−] ions can further promote the cubic phase formation and enhance the film crystallinity and orientational order as well. The role of Br[−] ion was also confirmed by Unold et al. that the presence of Br[−] ions in the films can stabilize the metastable perovskite phases upon cooling down.^[84] They further constructed phase diagrams for CsPb(Br_xI_{1−x})₃ containing α , β , γ , and δ -phase during heating and cooling processes by in situ GIWAXS and X-ray fluorescence (XRF) measurements. The substitution of large I[−] ions with smaller Br[−] ions can reduce the required transition temperature to the perovskite phase as well as the octahedral distortion during cooling down. Recently, our group employed in situ GIWAXS to study the working mechanism of light soaking induced performance enhancement of CsPb(I_{1−x}Br_x)₃-based PSCs and quantified the temperature increase under illumination based on the lattice expansion of the perovskite and reference materials.^[19] It was suggested that the light soaking effect was mainly contributed from the defect passivation of light activated halide migration and excess PbI₂ in the precursor can effectively suppress the light soaking effect.

Besides X-site engineering, further A-site engineering could enhance the cubic phase of all-inorganic perovskites as well as their device performance. These studies are usually based on CsPbI₂Br, which is one of the most used mixing halide compositions due to the relatively stable cubic phase and a suitable bandgap of 1.9 eV.^[85] It was found that the formation of the CsPbI₂Br cubic phase was advanced and strengthened by a trace of organic GA⁺ cations.^[86] Three crystallization processes

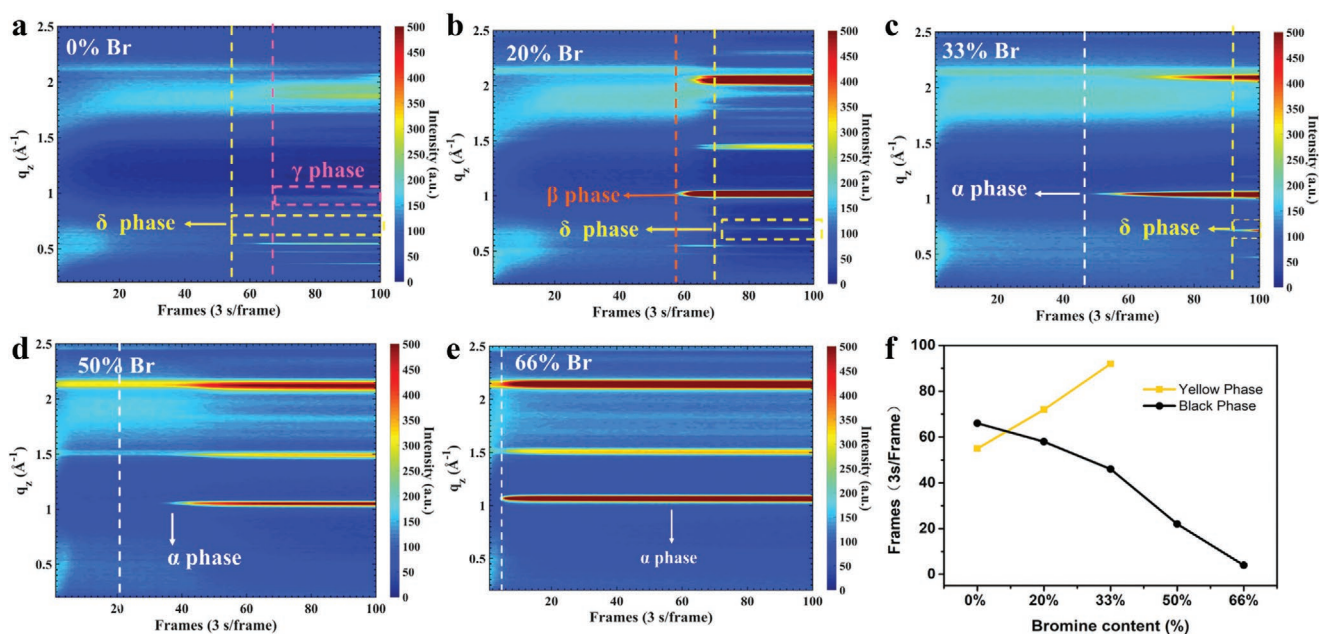


Figure 18. Time-resolved GIWAXS profiles for a) CsPbI₃, b) CsPb(I_{0.8}Br_{0.2})₃, c) CsPb(I_{0.67}Br_{0.33})₃, d) CsPb(I_{0.5}Br_{0.5})₃, and e) CsPb(I_{0.34}Br_{0.66})₃. f) Emergence time of the perovskite black phase and photo-inactive yellow phase in CsPb(I_{1−x}Br_x)₃ (x = 0, 0.2, 0.33, 0.5, 0.66) films. Reproduced with permission.^[83] Copyright 2021, Elsevier.

during the film formation of CsPbI₂Br were successfully identified: I) the precursor stage; II) the perovskite nucleation and growth stage; III) the crystallinity saturation stage. The GA⁺ can apparently facilitate the cubic CsPbI₂Br crystal nucleation and growth processes. Moreover, the phase transition to the yellow phase during stage III was greatly suppressed, delivering a better film morphology and higher device efficiencies over 14%.

Similar to organic–inorganic perovskite systems, the crystallization process of all-inorganic perovskites is also influenced by the deposition procedure. In comparison to the spin-coating

process, the blade-coating method can deliver a distinct film formation process of all-inorganic perovskites. Zhao et al. carried out in situ GIWAXS characterizations to explore the film formation dynamics of CsPbI₂Br during blade coating under different processing temperatures, as shown in Figure 19.^[28] At room temperature (25 °C), the as-cast film dried very slowly and maintained at the disordered sol–gel state (Figure 19a), while an elevated temperature of 80 °C allowed the fast crystallization from the sol–gel precursor to the perovskite phase (Figure 19b). Intriguingly, a sequential crystallization from CsPbBr₃ to CsPbI₂Br was

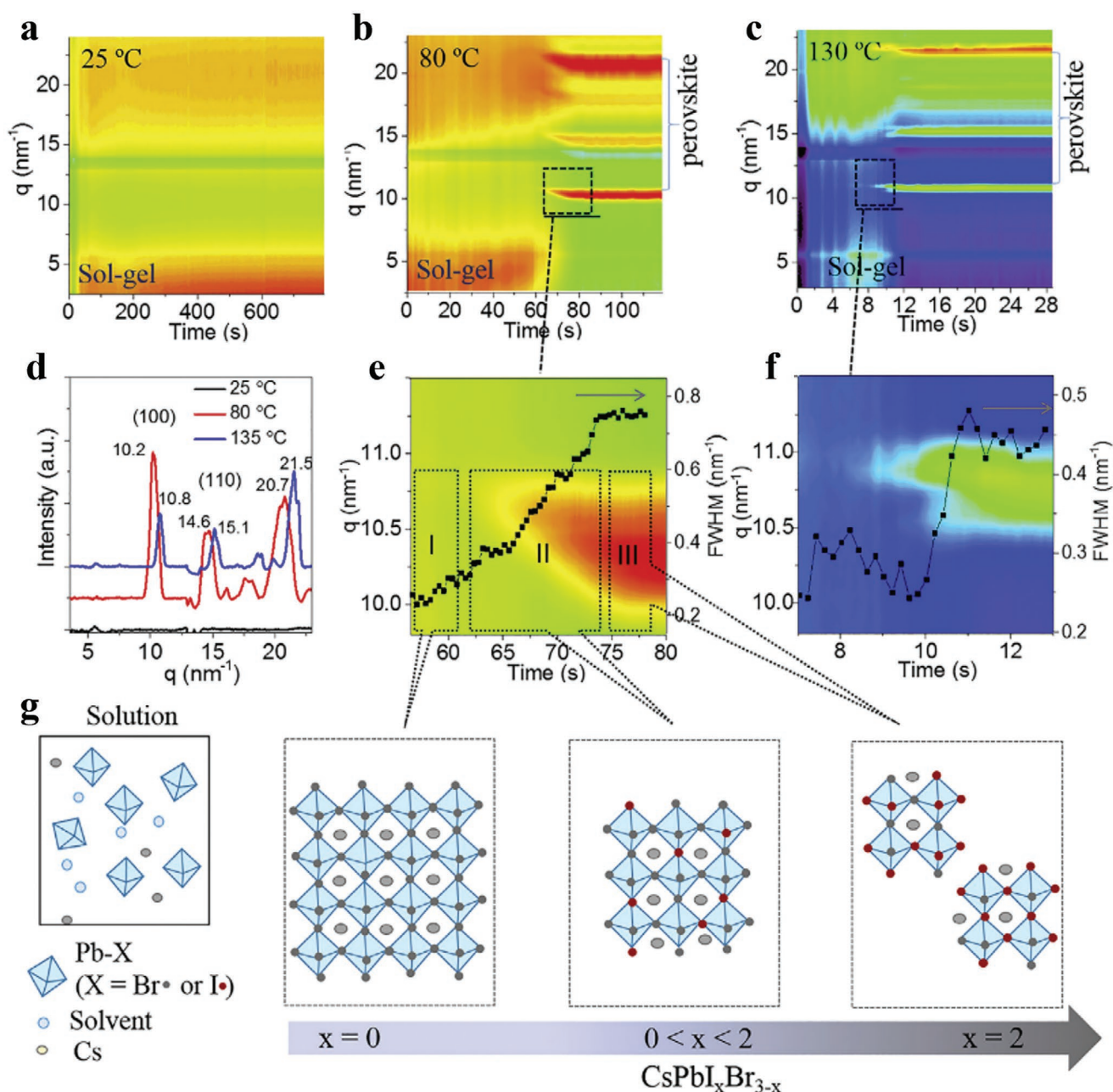


Figure 19. Structural evolution of the all-inorganic CsPbI₂Br perovskite. In situ GIWAXS measurements during blade-coating processes at a) 25 °C, b) 80 °C, and c) 130 °C. d) The corresponding intensity profiles of the last GIWAXS patterns taken at the end of the blading. The evolutions of the (100) perovskite diffraction peak and the FWHM values during blade coating at e) 80 °C and f) 130 °C. g) Schematic illustration of the formation of the CsPbI₂Br perovskite at 80 °C. Reproduced with permission.^[28] Copyright 2019, Elsevier.

observed at 80 °C during the film formation in Figure 19e, which was a vital process because the enhanced mass transport was in favor of the high-quality crystallization of the film, as illustrated in Figure 19g. However, the temperature at 130 °C was too high to give time for the high-quality sequential reaction (Figure 19c,f), as confirmed by the low crystallinity and poor morphology. As a result, highly crystalline, uniform, and pinhole-free CsPbI₂Br films were obtained, giving rise to a high PCE of 14.7%.

In brief, the crystallization process of all-inorganic perovskites does not involve the aforementioned organic-related intermediate phases that are commonly observed in the organic–inorganic perovskites.^[87] Instead, the phase transition mainly occurs between orthorhombic δ -CsPbI₃, orthorhombic γ -CsPbI₃, tetragonal β -CsPbI₃, and cubic α -CsPbI₃, among which γ , β , and α -CsPbI₃ are photo-active black phases but metastable at room temperature while δ -CsPbI₃ is the photo-inactive yellow phase but the most stable due to the lowest formation energy.^[13] Therefore, promoting the formation of black phases and suppressing the yellow phase during the film formation process are essential for achieving high-quality all-inorganic perovskite films. Smaller halide ions Br[−] and Cl[−] can both facilitate the formation of black phases and inhibit the spontaneous phase transition to the yellow phase, owing to the higher electronegativity.^[83] In the meantime, large cations GA⁺ can also achieve the same goal because of the stronger hydrogen bonding.^[86] Another interesting finding is the sequential crystallization from CsPbBr₃ to CsPbI₂Br during the formation of CsPbI₂Br perovskite films by employing the blade-coating method, realizing high crystallinity, uniform, and pinhole-free films.

3.5. Long-Chain Ammonium Incorporated 2D Perovskites

To improve the moisture stability of PSCs, hydrophobic long-chain ammonium cations such as phenylethylammonium (PEA⁺)^[88] or butylammonium (BA⁺)^[12] were introduced to divide the bulk perovskite frameworks into slabs, forming 2D perovskites.^[89] The general formula of 2D perovskites is (A')_m(A)_{n−1}B_nX_{3n+3}, where A' can be divalent or monovalent long-chain ammonium bulky cations, and A can be small cations such as MA⁺, FA⁺, and Cs⁺. In this part, we will mainly focus on the MA-containing 2D perovskites due to the superior optoelectronic properties of MA-containing 2D perovskites and limited studies of in situ GIWAXS on other 2D perovskites.

Owing to the hydrophobic properties of the long-chain ammonium cations, the long-term stability of the 2D PSCs under constant illumination and high humidity levels was significantly enhanced.^[12] The formation of 2D perovskites typically requires a high processing temperature or additives during spinning. Otherwise, the film coated at room temperature shows a 3D-like random orientation (Figure 20), indicating the crystal structure of 2D perovskites is critically dependent on the fabrication procedure. Liu et al., therefore, investigated the underlying crystallization mechanism of 2D BA₂MA₃Pb₄I₁₃ perovskites with the help of in situ GIWAXS characterization.^[90] All the films formed at room temperature comprised solvate complexes and PbI₂ crystals regardless of the choice of perovskite solvents (Figure 21a,b), giving rise to the

formation of multiple RP phases with random orientation. By contrast, direct crystallization of the BA₂MA₃Pb₄I₁₃ RP phase without intermediate phases was observed under a high processing temperature of 90 °C owing to the accelerated intercalation of ions and decreased nucleation barrier, as presented in Figure 21c. The hot-casting process can deliver high-quality 2D perovskite films with a preferential orientation, high phase purity, and improved optoelectronic properties. The resulting PSCs exhibited high efficiencies over 12%. Besides the deposition temperature, the underlying substrate is also crucial for influencing the crystal orientation. Siffalovic et al. observed an isotropic crystallization process of 2D perovskites on an FTO substrate by in situ GIWAXS.^[91] In contrast, a strong preferential orientation can be induced by employing a graphene substrate, which provides a π – π interaction with perovskites and thus in turn leads to a preferential orientation. Choi et al. further elucidated the formation mechanism of vertically oriented 2D BA₂MA₃Pb₄I₁₃ perovskites in which the nucleation and growth occurred from the liquid–air interface rather than the substrate–liquid interface.^[92] It was found that 2D perovskites tend to nucleate and grow at the liquid–air interface because of the strong surface tension of the perovskite precursor solution.^[93] In the light of this finding, Kanatzidis et al. unveiled three stages in the film formation of 2D perovskites during one-step spin coating: sol–gel, oriented 3D, and 2D.^[24] The sol–gel stage comprised 1D DMF-coordinated phases, which then transform to oriented 3D perovskites at the air–liquid interface, followed by the subsequent nucleation forming 2D perovskites eventually.

Based on the understanding of the crystallization kinetics of the 2D BA₂MA₃Pb₄I₁₃ perovskites, several approaches were further explored to regulate the film formation process such as inorganic cations doping. As shown in Figure 22a,b, Cs⁺ ions were reported to prolong the crystallization duration of 2D perovskites,^[94] which was beneficial for larger grains and better crystallinity. More interestingly, the doping of K⁺ formed desired vertically oriented 2D BA₂MA₃Pb₄I₁₃ perovskites without the need of a high temperature (Figure 22b,c), and the trap states in the film were reduced as well.^[95] As a result, the PCE of 10% K⁺-doped 2D PSCs was improved from 8.46% to 11.3%.

Besides BA⁺ cations, other ammonium ions (e.g., PEA⁺, guanidinium cation (GA⁺)) were also employed to construct 2D perovskites, exhibiting distinct crystallization kinetics. Sargent et al. monitored the RP phase formation with PEA⁺ ions during the spinning process using in situ GIWAXS.^[96] The film experienced a sequential conversion from the precursor to solvent complexes to RP phases, and they found that the decreasing rate of solvent complex peaks mirrored the growth rate of the perovskite peak during annealing, indicating a direct phase transition from solvent complexes to RP phases. Specifically, the large solvent complexes play an important role to provide a framework to facilitate the construction of PEA-based 2D perovskites. Zhao et al. also reported that the intermediate solvate complexes can serve as a scaffold for the growth of GA-based 2D perovskites.^[97] They undertook in situ GIWAXS studies on the crystallization process of 2D perovskite GAMA₃Pb₃I₁₀, and identified a key intermediate phase GA₂PbI₄ that determines the hierarchical phase distribution from low-*n*

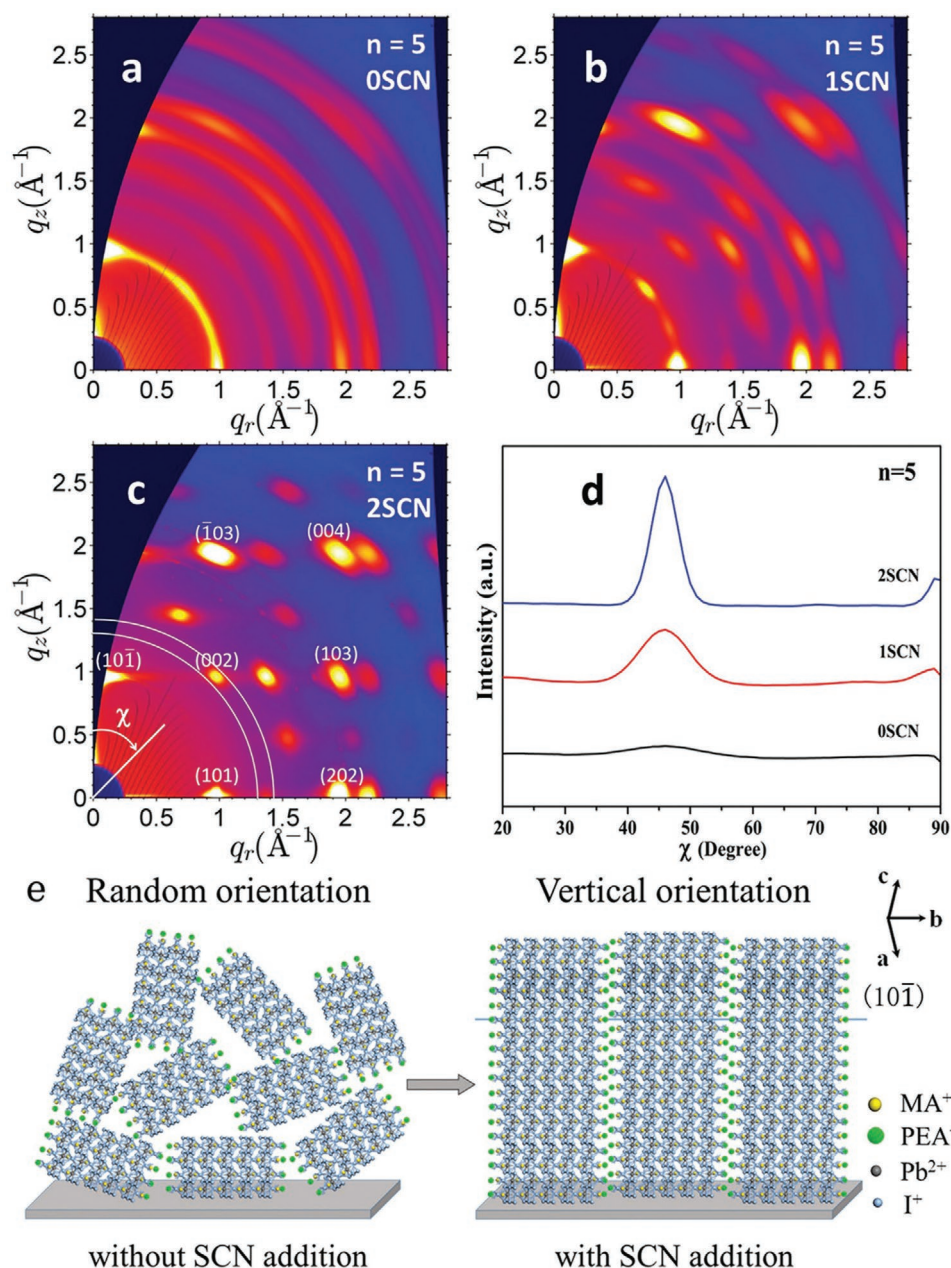


Figure 20. 2D GIWAXS patterns of (PEA)₂(MA)₄Pb₅I₁₆ perovskite films with a) 0SCN, b) 1SCN, and c) 2SCN. d) The corresponding polar intensity profiles and e) schematic interpretation from random orientation to vertical orientation. Reproduced with permission.^[88] Copyright 2018, Wiley-VCH.

to high- n in the antisolvent-processed film. Together with transient absorption (TA) spectra, a spontaneous charge transfer between the RDP phases with different n values was confirmed, resulting in an improvement of the PCE to over 14%.

Table 3 summarizes the detailed phase transition pathways for different 2D perovskites under various deposition procedures. In summary, compared with 3D perovskites, the formation of BA-based 2D perovskites typically requires the one-step hot-casting process which promotes a direct phase transition from the precursor to 2D perovskites without the appearance of the intermediate phase.^[92] Moreover, the role of alkali cations in the 2D perovskite crystallization kinetics is quite different from that in the 3D case. Cs⁺ can slow down the crystallization of

2D perovskites and consequently improve the film quality;^[94] K⁺ can facilitate the formation of 2D perovskites even at room temperature and regulate the crystal orientational order.^[95] As for PEA- and GA-based 2D perovskites, however, intermediate phases are believed to serve as critical scaffolds for the further formation of 2D perovskites,^[96,97] suggesting the dependence of the film formation processes on the ammonium cations.

4. Crystallization Kinetics of Lead-Free Perovskites

With huge improvements in the device efficiency and stability of Pb-based PSCs, the toxicity of lead has been an inevitable

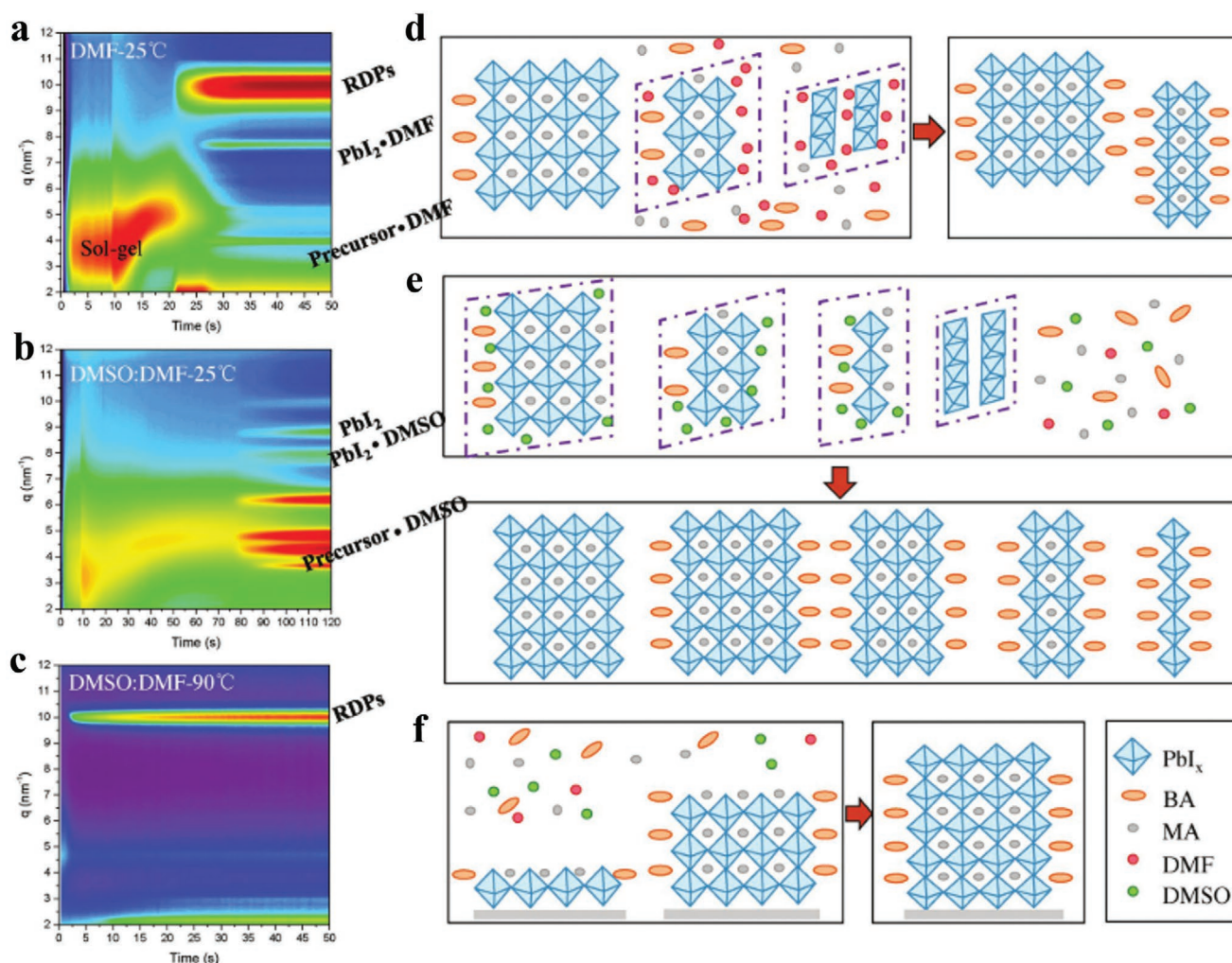


Figure 21. a–c) In situ GIWAXS measurements of 2D $\text{BA}_2\text{MA}_3\text{Pb}_4\text{I}_{13}$ perovskite films under different fabrication conditions and d–f) the corresponding schematic models of the film formation processes. Reproduced with permission.^[90] Copyright 2018, Wiley-VCH.

hurdle that will eventually impede the commercialization of PSCs.^[10,98] It is highly demanded to develop environmental-friendly PSCs based on nontoxic elements such as tin (Sn),^[99] germanium (Ge),^[100] bismuth (Bi),^[101] Among them, Sn-based PSCs have attracted the most attention owing to a more suitable bandgap and higher charge mobility. However, the efficiency of Sn-based PSCs is still lagging far behind the Pb-based counterpart.^[102] One of the main reasons is that Sn^{2+} is easily oxidized to Sn^{4+} upon exposure to air, which is equivalent to p-type doping. The excessive Sn^{4+} affects the carrier mobility and increases the recombination of electron–hole pairs, thus significantly reducing the V_{OC} of the device.^[103] Fortunately, the oxidation of Sn^{2+} can be effectively suppressed with the addition of additives (i.e., SnF_2),^[104] large organic cations (i.e., GA^+ , PEA^+),^[105] B-site alloying (i.e., Ge^{2+}),^[106] tin powder,^[107] etc.

Another critical issue of Sn-based perovskites is the fast crystallization during the film formation process because of its low formation energy. The fast crystallization makes it tricky to precisely control the film morphology and to achieve smooth, uniform, and pinhole-free films, being detrimental to

the device performance.^[109] With the aid of in situ GIWAXS characterizations, the underlying crystallization mechanisms of tin-based perovskites were also investigated, providing insights for the control of the film formation process. For instance, Portale et al. analyzed the crystallization kinetics of FASnI_3 by in situ GIWAXS, and observed a direct phase transition from the disordered precursors to FASnI_3 perovskite crystals at room temperature (Figure 23a,b), without the formation of any intermediate or complex phases.^[108] However, the fast crystallization leads to a discontinuous film with poor morphology, as shown in Figure 23c. Interestingly, incorporating PEA^+ molecules can induce the formation of 2D RP perovskites during the crystallization process, giving rise to the growth of a highly oriented top layer. The RP perovskites can modify the film morphology and thus deliver a high efficiency of $\sim 9\%$. It was indicated that the device performance of Sn-based perovskites is sensitive to the crystallization kinetics and the control of the crystallization rate is critical to the formation of high-quality films. Besides the Sn-based perovskites, the direct phase transition from precursor-to-perovskite was also found in another lead-free

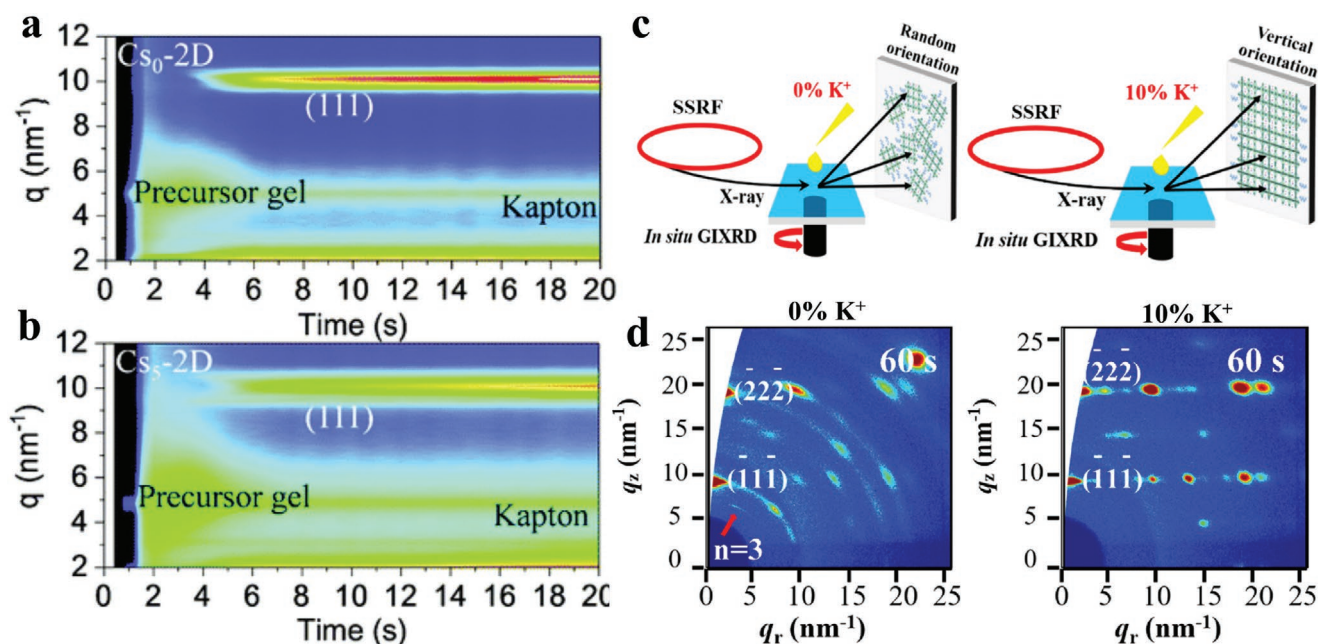


Figure 22. BA-based 2D perovskite crystallization with in-organic cations. In situ GIWAXS results of 2D perovskites a) without and b) Cs^+ doping. Reproduced with permission.^[94] Copyright 2017, Royal Society of Chemistry. c,d) Crystallization processes for 2D perovskites without and with K^+ . Reproduced with permission.^[95] Copyright 2020, American Chemical Society.

perovskite—methylammonium bismuth iodide ($\text{MA}_3\text{Bi}_2\text{I}_9$) by Amassian et al. They also revealed that the antisolvent dripping amid the crystallization process is crucial to fabricating continuous films of $\text{MA}_3\text{Bi}_2\text{I}_9$ and functional photovoltaic devices.^[110]

Although some efforts have been devoted to controlling the lead-free perovskite crystallization process, the underlying formation mechanisms remain unclear on account of very limited studies on the crystallization kinetics of lead-free perovskites. Based on existing in situ GIWAXS results, it was found that lead-free perovskites generally experience a fast and direct phase transition from the precursors to the perovskite phases even at room temperature,^[108] which is quite different from those of the Pb-based counterparts that generally involve some intermediate phases during the film formation process.^[14a] Nevertheless, it is challenging to further improve the Sn-based perovskite film morphology due to the uncontrollable crystallization. Although some approaches have been reported to regulate the crystallization kinetics,^[111] the mysterious phase transition pathways remain to be uncovered by in situ GIWAXS characterization.

5. Summary and Outlook

In conclusion, we have systematically reviewed various perovskite crystallization processes revealed by in situ GIWAXS and summarized the proposed film formation mechanisms of different perovskite systems, including 3D organic–inorganic perovskites, 2D perovskites, all-inorganic perovskites, and lead-free perovskites. Specifically, MA-based 3D perovskites generally experience a phase transition from precursor to intermediate phases to the perovskite phase, while the formation of FA-based 3D perovskites evolves from the yellow phase (δ -phase) instead of intermediate phases. The formation of 2D

perovskites can involve several intermediate phases, whereas the hot-casting treatment leads to a direct phase transition from the precursor state to the perovskite phase. In comparison, the crystallization of all-inorganic perovskites may lead to photo-inactive δ , and photo-active γ , β , and α - CsPbI_3 phases, and the incorporation of Br^- and Cl^- ions not only promotes the formation of the photo-active phases but also suppresses the formation of the δ -phase. As for lead-free perovskites, only direct phase transitions from precursor to the perovskites phase were observed without any intermediate or secondary phases, which may be ascribed to the low formation energy.

Although extensive research has been carried out on the perovskite crystallization kinetics, there remain unsolved puzzles, especially for 2D and lead-free perovskites, which are expected to be uncovered by in situ GIWAXS measurements. During the formation of 2D perovskites, only intermediate phases and oriented perovskite structure can be observed, while the structural influence of long-chain ammonium ligands is still unclear during the crystallization process. Since the preferential orientation of Pb-I octahedra is induced by the ligands, understanding the transition process of the ligands with the aid of in situ GIWAXS is the key to reveal the formation mechanism of oriented perovskite crystals. Besides, studies of in situ GIWAXS on the crystallization process of lead-free perovskites are also very limited. It has been realized that the lead-free perovskites generally experience very fast crystallization and direct phase transition pathways have been reported for tin-based perovskite and double perovskite systems. However, the fast crystallization gives rise to poor film morphology, which is detrimental to the device performance. Therefore, a control over the crystallization rate of Sn-based perovskites is highly demanded, which requires further understandings of the crystallization pathways by virtue of in situ GIWAXS characterization.

Table 3. Phase transitions during the film formation process of 2D perovskites and the corresponding synchrotron beamlines.

Perovskite	Deposition procedure	Phase transition	Treatment or remark	Beamline	Ref.
BA-based 2D Perovskite	One-step spinning	Precursor → oriented 3D perovskite → 2D perovskite	The crystallization occurs from the liquid–air interface.	8-ID-E, ANL	[24,92]
		K ⁺ doping: precursor → 2D perovskite		BL14B1, SSRF	[95]
	One-step hot casting	Precursor → 2D perovskite	Cs ⁺ doping can prolong the crystallization kinetics.	D1, CHESS	[90,94]
PEA-based 2D Perovskite	One-step spinning	Precursor → intermediate phase → intermediate phase/2D perovskite		D1, CHESS	[96]
	Annealing	Intermediate phase → 2D perovskite		SSRL	[96]
GA-based 2D Perovskite	One-step spinning	Precursor → GA ₂ PbI ₄ intermediate → 2D perovskite		D1, CHESS	[97]

Furthermore, it is also worthwhile to further examine the degradation process of perovskites to figure out the origin of the instability by taking advantage of in situ GIWAXS technique. Different perovskite systems may possess disparate degradation mechanisms. For instance, the phase transition from the perovskite phase to the yellow phase in all-inorganic perovskites can

be accelerated by moisture, which can be reversed by heating. In contrast, the degradation of organic–inorganic perovskite induced by moisture is irreversible. Thus, moisture may play a different role in the degradation process of all-inorganic perovskites and organic–inorganic perovskites. It can be confirmed by in situ GIWAXS whether the moisture involves in perovskite

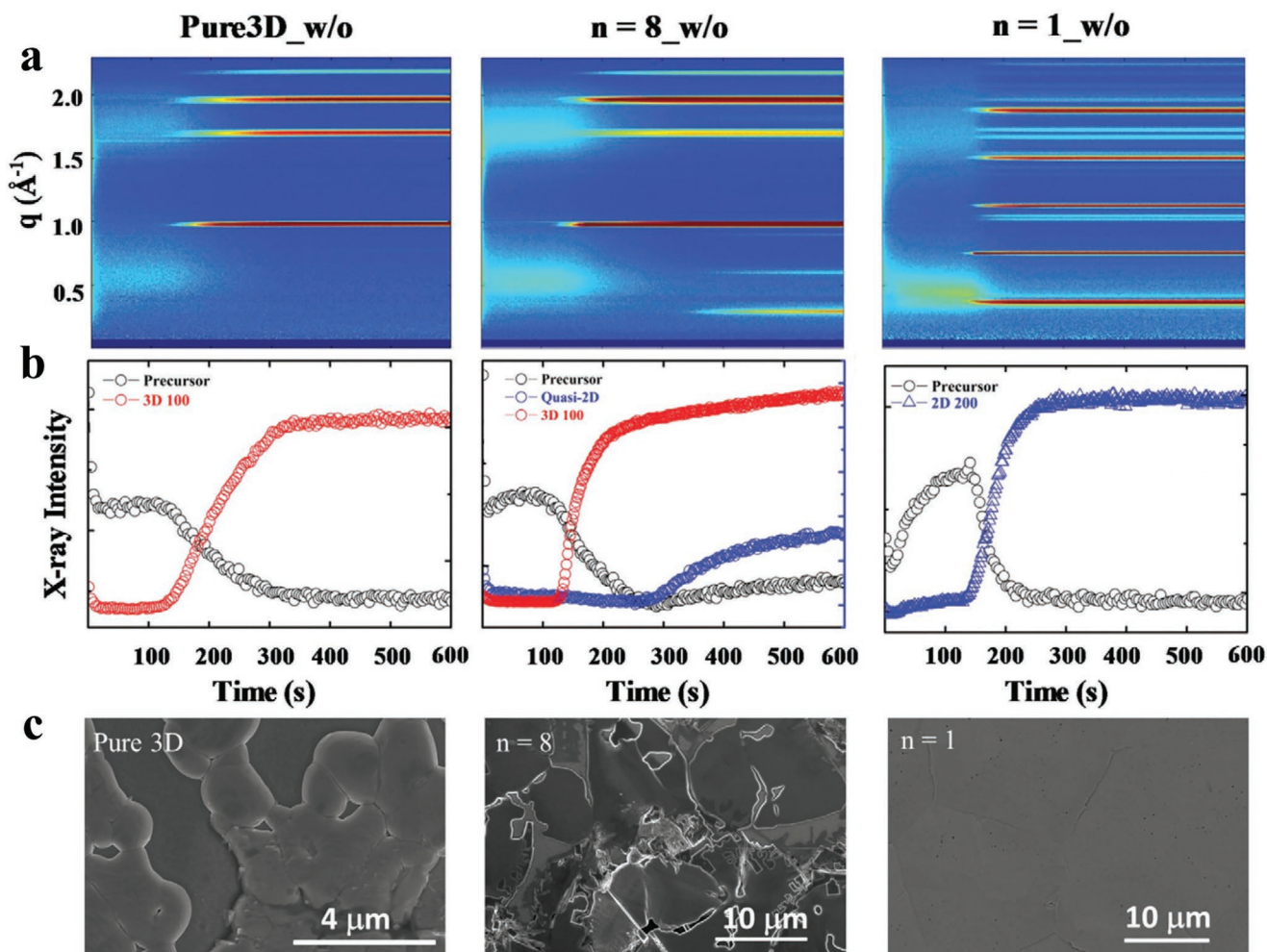


Figure 23. Crystallization kinetics of Sn-based perovskites. Time evolution of a) the integrated GIWAXS profiles, b) the integrated peak intensities, and c) the corresponding top-view SEM images for pure 3D FASnI₃, RP $n = 8$, and pure 2D $n = 1$ perovskite films. Reproduced with permission.^[108] Copyright 2020, Wiley-VCH.

phase transitions or just a stimulus to the degradation. Furthermore, impacts of other external stresses such as oxygen, heat, and illumination on the stability of perovskites are deserved to be identified separately.

Finally, another interesting and promising direction is to combine the in situ GIWAXS characterizations with other in situ characterization methods such as current density–voltage (J – V) measurements, UV–Vis absorption, photoluminescence (PL) measurements, in order to establish the direct correlation between the crystal structural evolution and the film/device properties. Besides, the development of these in situ characterization strategies will not only benefit the research of perovskite materials, but also readily is migrated to the study of other thin-film-based optoelectrical devices.

Besides in situ GIWAXS, it is worth noting that in situ grazing-incidence small-angle X-ray scattering (GISAXS) is another powerful technique to investigate the nanoscale structural evolution during the perovskite film formation process. The sample-to-detector distance for GISAXS measurements is usually several meters compared with the distance of about several centimeters in GIWAXS.^[112] Therefore, the probing length scale of GISAXS is usually between ≈ 1 and ≈ 100 nm, which can characterize the change of perovskite grain size during film formation and monitor the perovskite nucleation rate as well as crystallization kinetics.^[113] For instance, Lidzey et al. revealed an increase in the average height of $\text{CH}_3\text{NH}_3\text{PbI}_{3-x}\text{Cl}_x$ perovskite grains during thermal annealing by fitting the slope after the Yoneda peak in q_z from in situ GISAXS.^[50] Meanwhile, the shift of the Yoneda peak to a higher q_z value also implies an increase in the average density of the film. Nevertheless, Krebs et al. pointed out that the Yoneda peak shift from in situ GISAXS could be partially caused by the X-ray beam damage rather than the developing perovskite layer, since the Yoneda peak returned back to the original position after they moved the sample.^[114] Besides Yoneda peak shift, perovskite nucleation and growth processes were also studied by Dunbar et al. by fitting GISAXS data with a power-law model, demonstrating faster nucleation with the addition of HI and DIO additives.^[115] The slope of the power-law scattering curves was related to the surface fractal dimension.^[116] Lin et al., therefore, found that the surface roughness of perovskite films can become larger along with annealing owing to the increased surface fractal dimension, which explained the variation in resulted device performance.^[117] Despite these interesting studies, the investigation on the perovskite film formation process by in situ GISAXS is still limited and further exploration in this aspect should definitely be beneficial to understanding the perovskite film formation mechanism.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

3D and 2D perovskites, all-inorganic perovskites, in situ GIWAXS, lead-free perovskites, perovskite film formation process

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